Word Wizard

"Non-verbal individuals often rely on communication tools that are very limited and difficult to customize, as users often have different needs. Educators would benefit from a device that converts physical input to verbal communication that is portable and customizable."

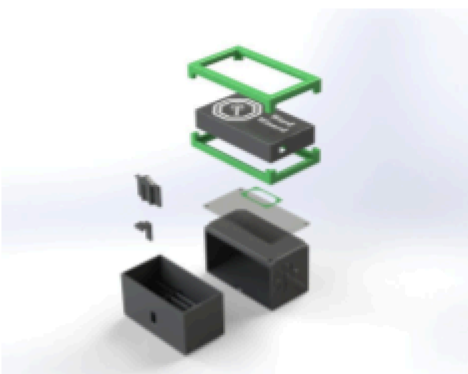


Figure 1: CAD Assembly of Device

Figure 2: Demonstration of use of device

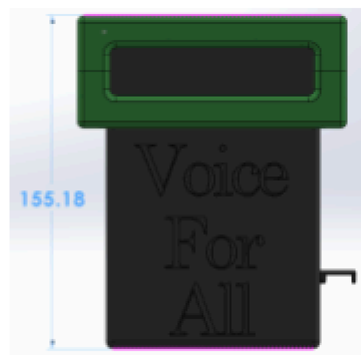


Figure 3: Dimensions of device from side (mm)

Figure 4: Dimensions of device from above (mm)

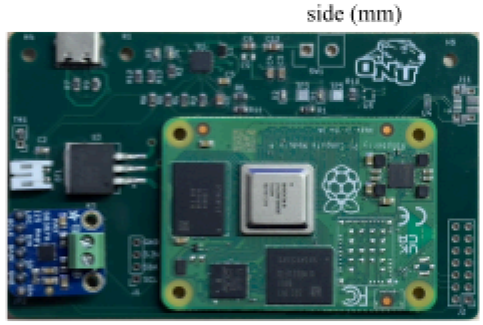


Figure 5: Electronics

Next Steps

While this design was extensively researched and revised over the course of the semester, our team recognized several features that could still be improved given we had more time. While our current storage system is innovative and easy to navigate, we believe the system could be improved by making the cards accessible from the top of the box rather than a drawer system to lower the complexity of operating the system for our primary stakeholder, the user. We also presume that making the device and the card system one piece would improve the manageability of Word Wizard for teachers and parents. Additionally we hope that in the future the bumper could be injection molded silicone so that we are not limited to 3-D printing capabilities. This is so we can make the bumper removable, similar to a phone case to improve the cleanability of the device. Finally we would implement an SPI communication protocol rather than the I2S protocol we currently use to optimize software speed for the device. These changes would help us compete in the global market more effectively and provide the best overall product for all stakeholders.

There are many ecodesign strategies we could employ in the further development of our product. We could optimize the end of life system by making all the hardware components accessible by common screws. This is so that we can increase the ease of disassembly and recyclability of the various different components. While this will be good for properly recycling components at end of life, it could tempt users to tinker and harm device functionality. Additionally we could optimize the initial lifetime by reducing the amount of soldered parts to connectors. While this makes repairability easier it could harm the reliability of the product as parts could be jostled loose during use. This is so that electronic parts can be easily swapped out to increase the lifespan of the device. Lastly we could further the reduction of materials by finding ways to further consolidate the footprint of the electronics. This will allow us to shrink the size of the device, reducing the necessary material required to fabricate all related parts. While this could reduce the footprint of the electronics, it could increase the complexity of the electronics further driving up costs. While these ecodesign strategies have pros and cons, we believe that there is a middle ground that makes these aspirations viable.

Broader Impacts

Table 1: Broader Impacts

	Benefit	Risks and Tradeoffs
People - Public Health, Safety & Welfare	- Utilizing a shock absorbent TPU bumper reduces injury risk when dropped or thrown - USB - C eliminates the need for proprietary application specific cables - Simple NFC tap input reduces cognitive load for the intended user	- Easy to access screws for easy repair can be a a safety hazard to young and curious individuals - TPU bumper increases overall footprint of device - Drawer system requires device to have two separate parts to manage
People - Global / Cultural Factors	- Color coded Fitzgerald Key is a widely recognized AAC standard - utilizing Piper TTS model allows the system to output multiple languages and dialects	- color coding may not transfer across different cultures - NFC programming requires a smartphone or other proprietary device
Planet - Environmental	- programmable reduces need for laminate and print waste - Modular electronic hardware allows for extended life of device, reducing waste - Rechargeable battery reduces single-use waste - ABS and TPU are recyclable materials	- Various electrical components like the PCB are non-biodegradable - Current manufacturing process is very energy intensive
Profit - Economic	- Lower cost than tablet alternative AAC's - Design is easily scalable and modular to accommodate large quantity needs for schools or care centers - Electronics are easy to fabricate and price gets lower as the bulk of the order increases	- a switch to SPI protocol would require further man-hours driving up our R&D costs - Our product has a long lifespan which will reduce our repeat business unless we come out with an updated more enticing product.

Appendix 1: Problem Framing

Overview: Problem Framing Documentation For FoD2 Semester Project 'Word Wizard'

Broader Impacts:

1. People - Inclusion and Accessibility to reduce stigma around disability
2. Planet - Long-lasting and purpose built solution allows for less material waste and power consumption
3. Profit - Lower cost and scalable solution compared to existing assistive technologies

Stakeholders:

Table 2: Stakeholders

	Stakeholder	*Impact*	*Design decision considerations*
1	Student	Usability, Comfort, Reliability	Safer casing, minimal simplistic inputs
2	Teachers	Integration and Manageability	Small, rechargeable, minimal design, with clear colors for locatability
3	Administrators	Budget and Policy Compliance	Low-tech design, Cost less than common alternatives (Tablet, or other repurposed smart device)
4	Parents	Independence and Safety	Soft outer casing, simplistic usability for a wide age range

Research:

Table 3: Preliminary Research

	Prompting Question	*Source*	*Specific information gathered*	*What will you do with this info?*
DR1				
1	How tall are classroom desks and tables?	[1]	24”-30”	This data will influence our durability benchmarks.
2	How can we smooth 3D prints?	[2, 3]	ABS can be smoothed with acetone, TPU can be smoothed with DMSO	Consider solvent smoothing for the surface finish of 3D printed parts

3	What is the most common consumer charging cable?	[4]	USB Type-C is "Common on modern laptops, smartphones, and game consoles. [It] is the new industry standard."	We will use a USB-C port for power input
4	What kind of 3D printing filament is the most durable?	[5]	ABS has a higher impact resistance than both PLA and PETG but it has worse printability.	ABS will be the preferred filament as it is the strongest and we will have to be more careful when printing with it.
5	Why use NFC tags instead of anything else for Voice recording?	[6]	NFC tags offer an easy, touch-based way to play, record, or update audio, making them highly accessible for tasks like labeling objects for the visually impaired or creating personalized voice messages without using complex menus.	We will use NFC tags for easy use and easy updating
6	What are the issues with paper communication cards?	Christina Teevan, Ashland Special Needs Ministry Executive Director	The commonly used paper cards are not durable, difficult to organize, and take concentration by both parties to communicate.	We will make a device to combine physical communication cards and text-to-speech communication devices.
DR2				
1	What is an effective dB level for sufficient communication and how will our enclosure affect that?	[1]	- Target output at listener: 60-65 dB - Upper practical range: 65-70 dB (if classroom noise is higher) This aligns with: - Normal speech = 60 dB - Slightly raised voice = 65–70 dB	Aim for a noise range between 60 -65 Decibels

2	How do educators currently store, organize, and manage communication cards in classroom environments?	[2]	Bins and Binder Pockets - Educators typically use baseball card organizers to track specific cards, and store larger amounts of cards in small storage bins	Develop a system that blends these two tracking systems. I.e. organized drawer
DR 3				
1	How can we reduce rattling of the speaker when mounted in the casing?	[1]	"When a speaker is mounted behind a cover, the sealing between the speaker and the surface dictate the Sound Pressure Level"[1] Air leaking out the speakers rim	Add a sort of gasket to seal the speaker
2	How can we further make cards easier to organize?	[2]	- Augmentative and Alternative Communication(AAC) systems [2] - Fitzgerald Key: Assigns colors to differentiate various grammatical categories [2] - Color coded designs require less cognitive load [3]	Color code the cards to better organize and track them

Assumptions:

1. Because classroom desks and tables are typically 24”-30” tall [1], we assume accidental drops from approximately 3 feet onto hard flooring are likely. Therefore, we will design the enclosure to withstand at least a 3 foot drop without functional failure.
2. Because some non-verbal individuals may also experience visual processing challenges, we assume high-contrast visual cues will improve usability. Therefore, we will use contrasting colors and clear iconography to indicate interaction zones.
3. Based on classroom environments where many interactable objects are used simultaneously, we assume communication cards may frequently be misplaced. Therefore, we will research and integrate a dedicated card storage or tethering solution.
4. Based on hygiene practices in schools and therapy settings, we assume the device will require frequent surface cleaning. Therefore, we will select enclosure materials (ABS/TPU/PLA) that tolerate repeated wiping without degradation. It also may be necessary to seal surface pores to decrease bacterial buildup.
5. Because USB-C charging cables are commonly included with consumer electronics [4] and are easily purchasable, we assume that the users will have access to these cables more readily than other types. Therefore, we will include a USB-C port for power input.

Design Specifications:

Table 4: Evaluation Metric

	Evaluation Metric	*Qty. to Measure*	*Units*	*Objective*	*Prototype Result*
1	Battery Life	Both	Watt-Hour (Wh)	≥ 30 Watt-hours	Utilize a rechargeable battery that still accounts for safety
2	Boot Time	EM	Seconds (s)	minimize	Optimize hardware utilizing a compute module
3	Response Time	EM	Seconds (s)	minimize	Optimize hardware utilizing a compute module
4	Sound Level	EM	Decibels (dB)	≥ 60	Functionally test different speaker grills for housing.
5	Portability	EM	grams(g)	minimize	Minimized unnecessary additional cables and hardware to make device as transportable as possible
6.	Variability of Word List	C	# of cards	≥ 30	Diversified vocabulary so device can be versatile in many scenarios
7.	Customizability of Word List	EM	# of ways to change each card	maximize	Cards can be changed on a need to need basis very easily through a simple smartphone

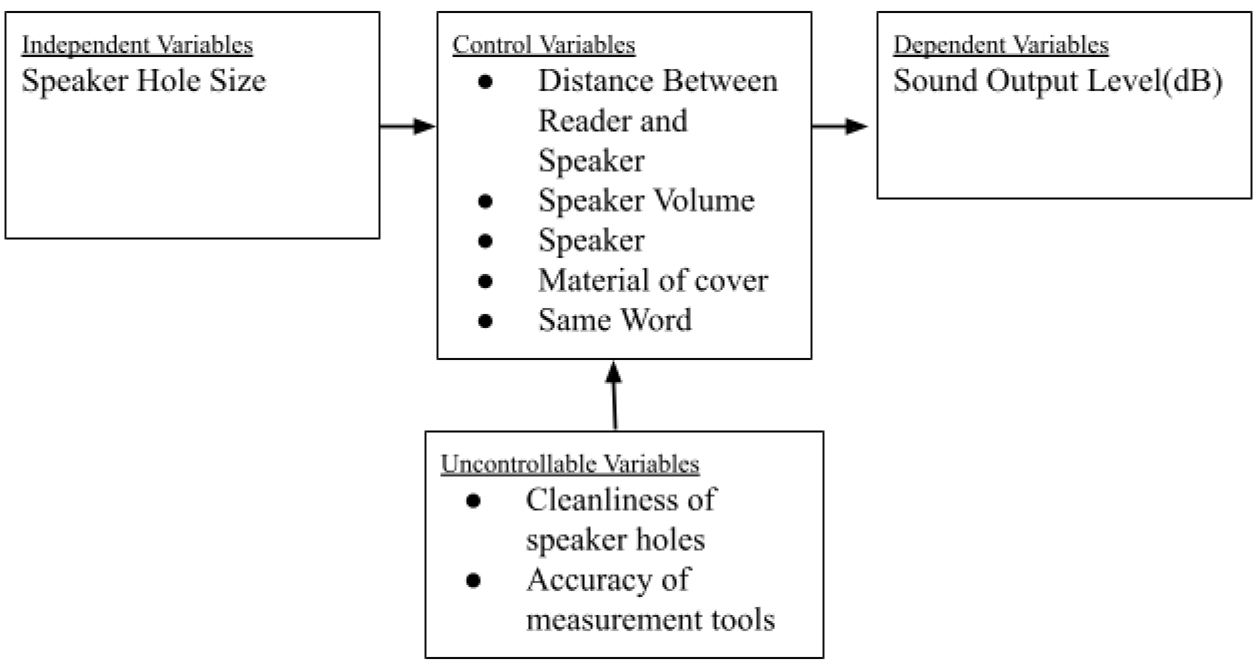
Appendix 2: Test Results

Test Title: Decibel Output Testing for Variable Speaker Holes

Purpose: To determine the best speaker cover to ensure the optimal sound clarity and quality determined by decibel levels.

Equipment:
- Speaker
- Microcontroller
- Amplifier
- Speaker covers (6 covers)
- Sound Level Meter
- Tape Measure
- Means to record data

Variables:



Experimental Setup:

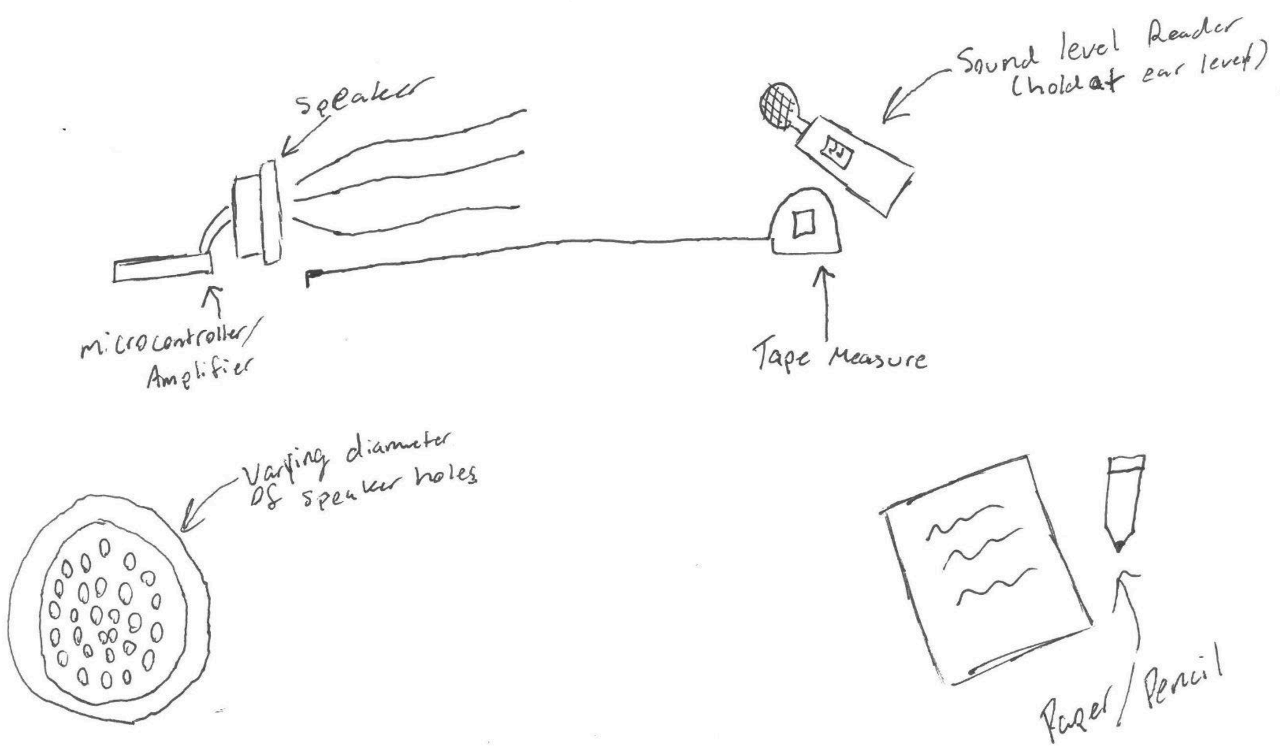
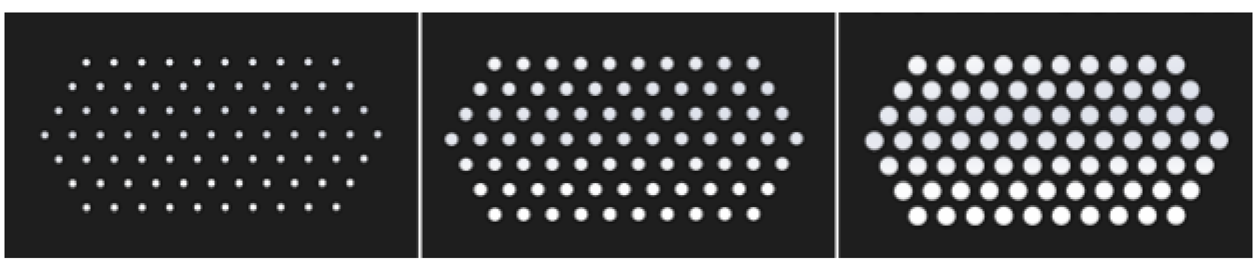


Figure 6: Setup to test audio loss due to diameter of holes



Figures 7-9: CAD renderings for varying speaker cover hole diameters (1.5mm, 2.5mm, 3.5mm)

Procedure:

1. Gather all materials.
2. Place materials on a table that is cleared and ready for the test.
3. Hook up the speaker to the amplifier + microcontroller apparatus.
4. Cover speaker with speaker cover
5. Hold sound level reader at ear level
    a. Speaker should be oriented towards you, still on the table
6. Use the sound level reader to collect data from the speaker output.
7. Output text to speech word from the speaker.
8. Repeat steps 4 through 6, changing / adding the speaker cover of varying hole diameters.
9. Write down results in the data table.
10. Write down any observations collected during the test.

Quantitative Results:

Table 5: Table to collect effective decibels across three tests at different amounts of obstruction

Decibels (dB)	Control (no cover)	Hole Diameters (mm)		
		D1	D2	D3
Run 1				
Run 2				
Run 3				
Average				

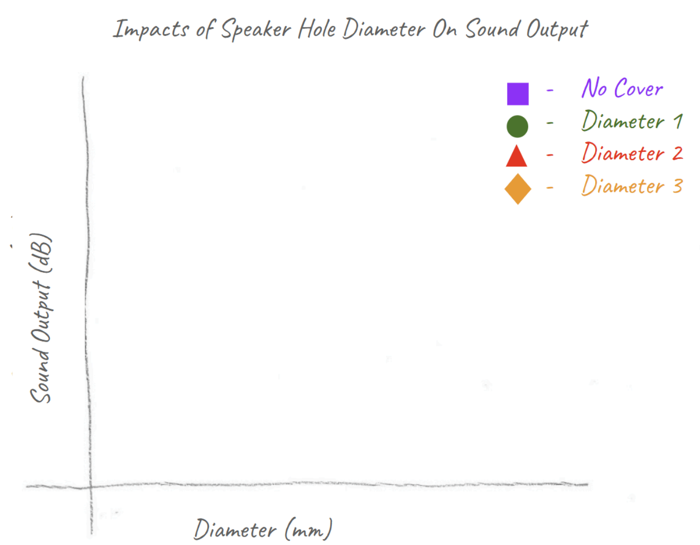


Figure 10: Scatterchart to compare effective decibels at different diameters of holes

Observations:

After conducting the experiment we have noted that the 2.5mm diameter speaker grill will give the best sound output. Along with the best sound output we observed that it will do better against keeping out dust and debris compared to its larger counterparts. The smaller diameters have too much of a sound loss to justify being better at keeping the dust and debris out.

Decibels (dB)		Control (no cover)	Hole Diameters (mm)					
			1.5	2.0	2.5	3.0	3.5	4.0
Phrases	"Why"	66.5	72.4	75.6	76.5	76.3	63.7	68.8
	"Please"	70.4	64.3	65.8	65.7	68.5	70.3	72.6
	"Stop"	68.6	58	59.4	72.9	57.3	75.6	63.1
	Average	68.5	64.9	66.9	71.7	67.4	69.9	68.2

Table 6: Raw test result data displaying test words utilized

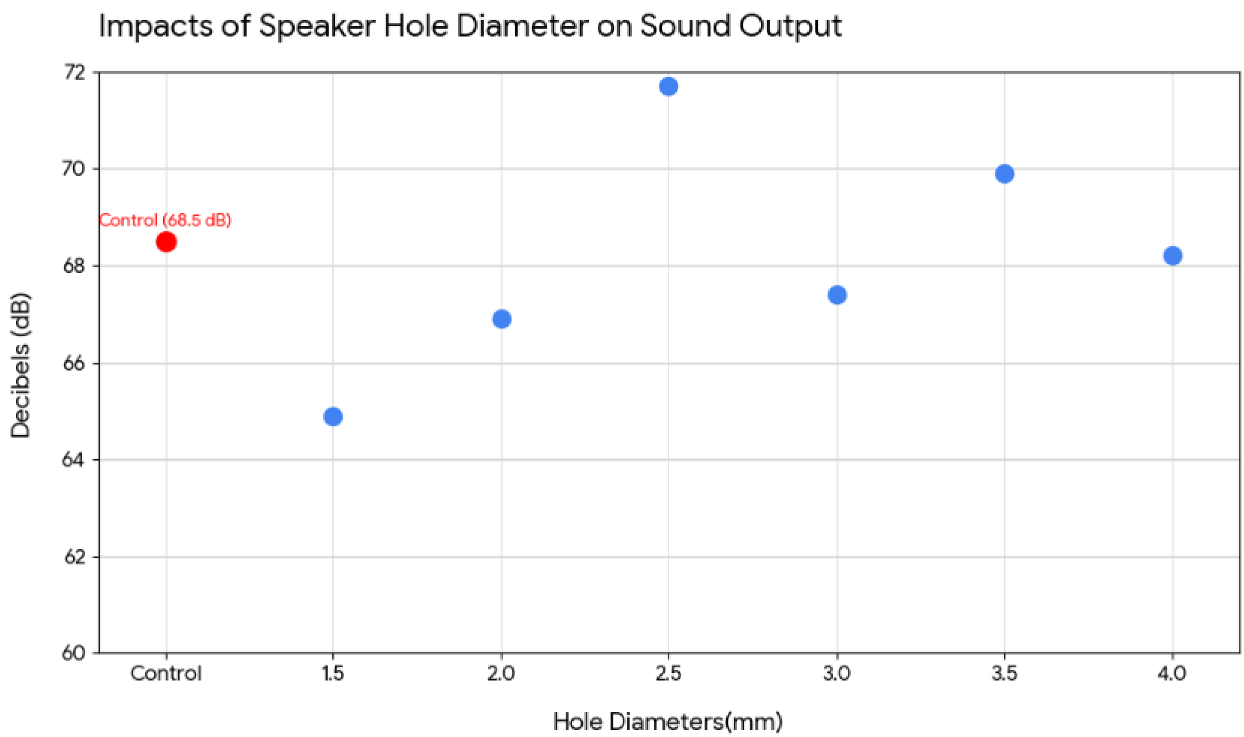
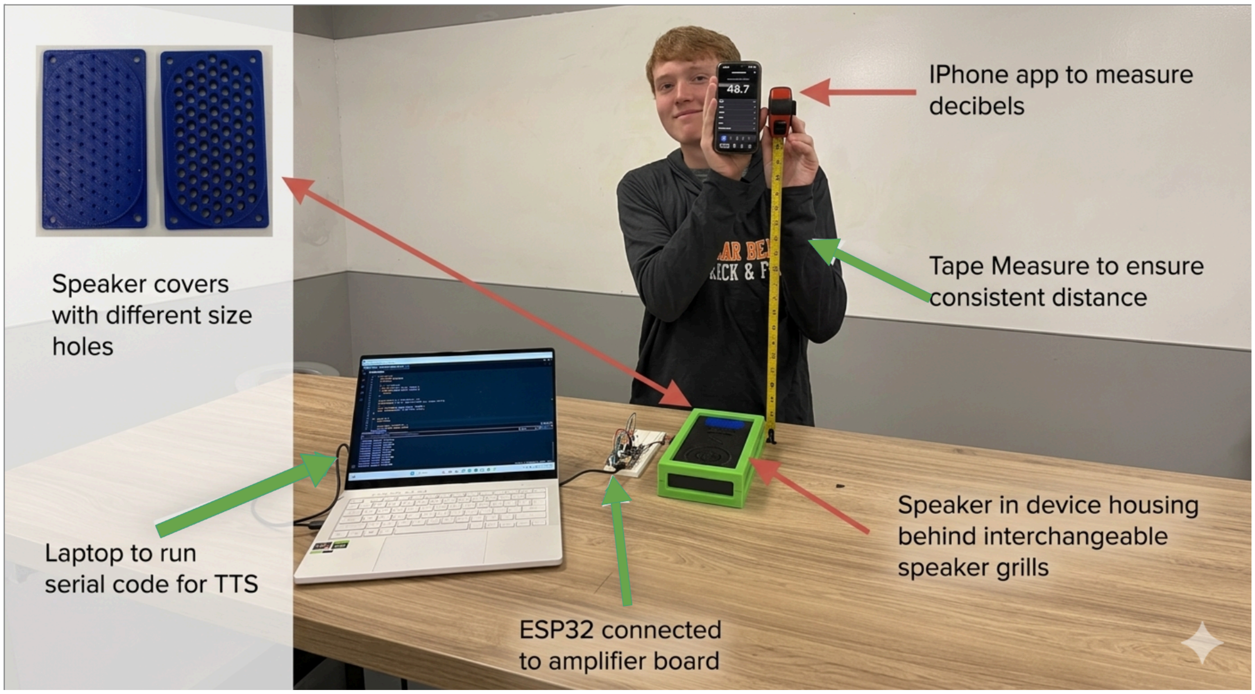


Figure 11: Plotted chart displaying decibels versus hole diameter data

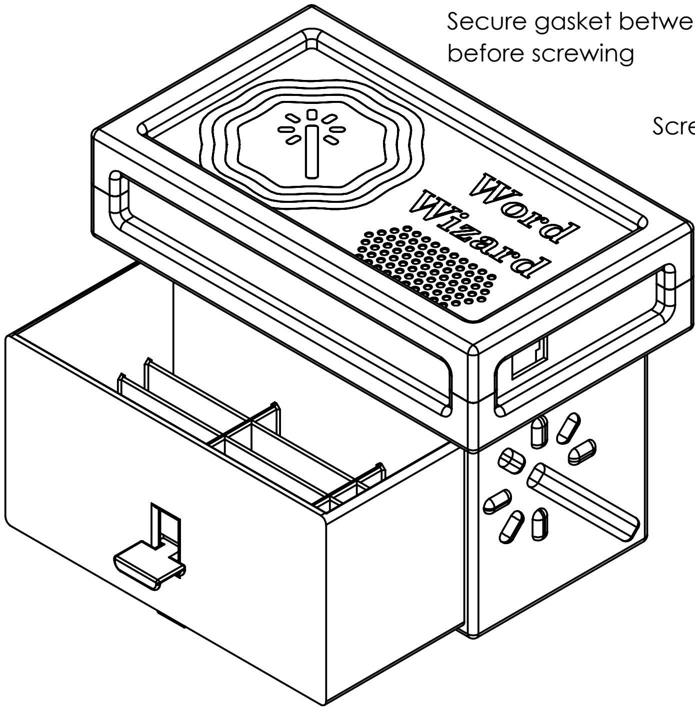
Picture:

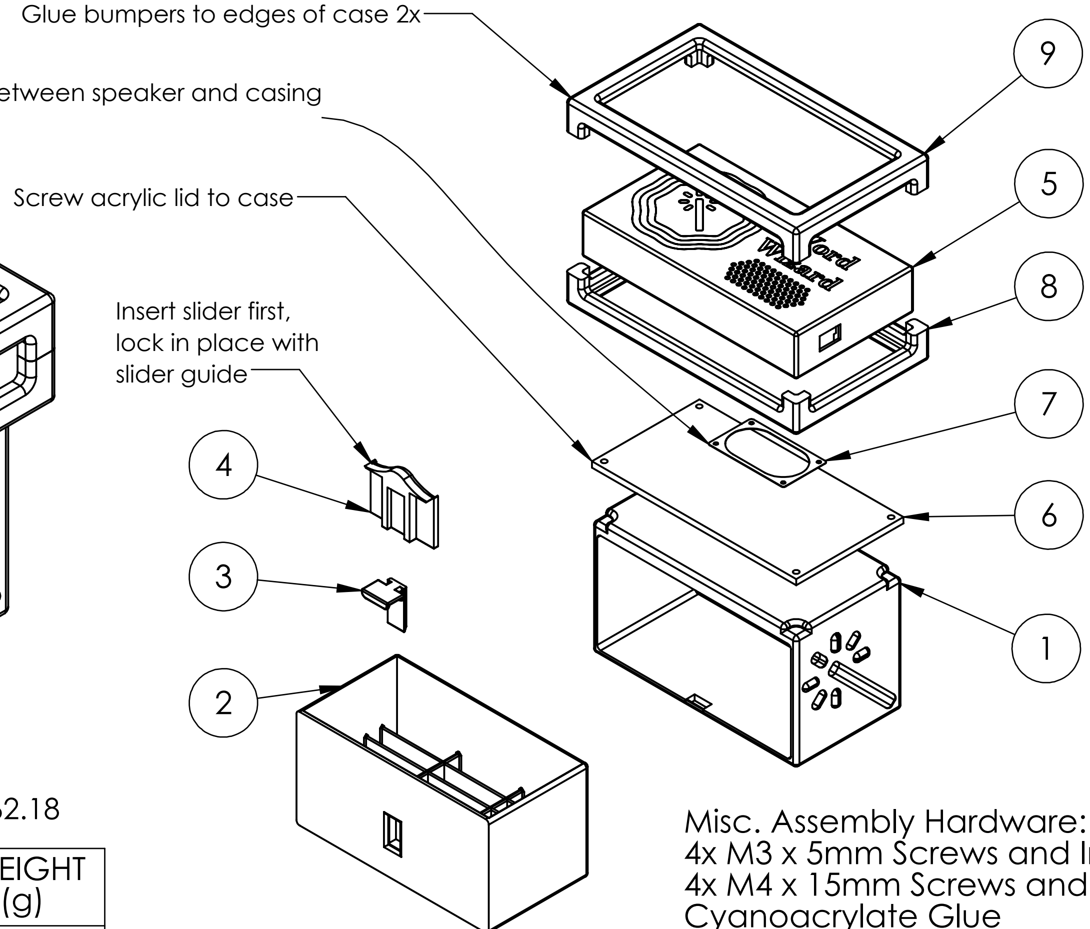


Conducted by: Ethan D'souza, Owen Willoby, Jeremiah Koenig, Aiden McManis

Date test conducted: 4/9/26

Appendix 3: Design Details





Overall Dimensions (LxWxH): 216.00 x 125.40 x 162.18

Misc. Assembly Hardware:
4x M3 x 5mm Screws and Inserts
4x M4 x 15mm Screws and Inserts
Cyanoacrylate Glue

ITEM NO.	PART NAME	MATERIAL	WEIGHT (g)
1	Drawer Box	PLA	237
2	Drawer	PLA	302
3	Slider	PLA	3.55
4	Slider Guide	PLA	9.58
5	Case	PLA	170
6	Lid	Cast Acrylic	118
7	Speaker Gasket	TPU	0.47
8	Bumper Bottom	TPU	52.4
9	Bumper Top	TPU	58.2

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL
SEE BOM

FINISH

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	EHD	4/19/26
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

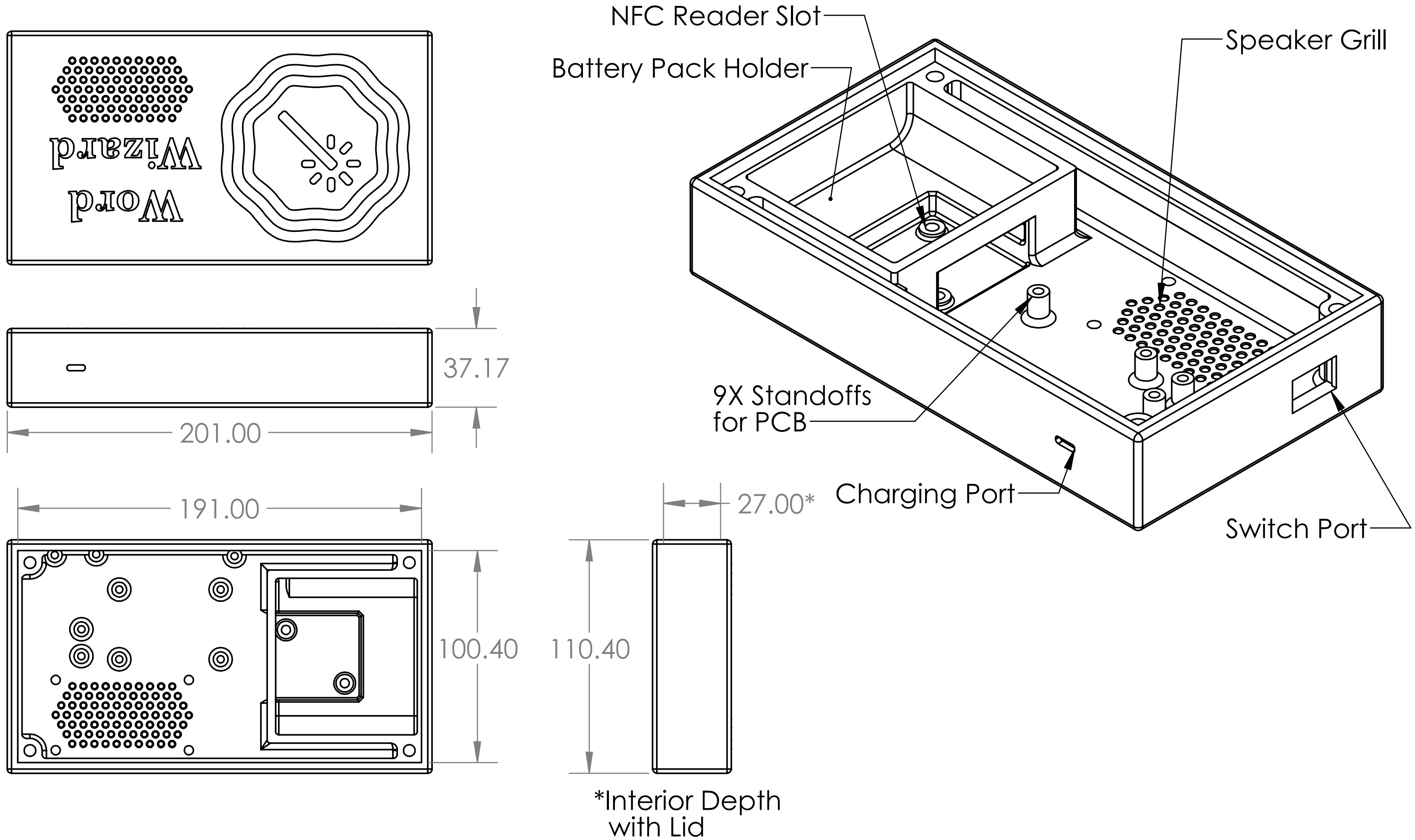
COMMENTS:

TITLE:

Word Wizard Assembly

SIZE: A | DWG. NO. | REV

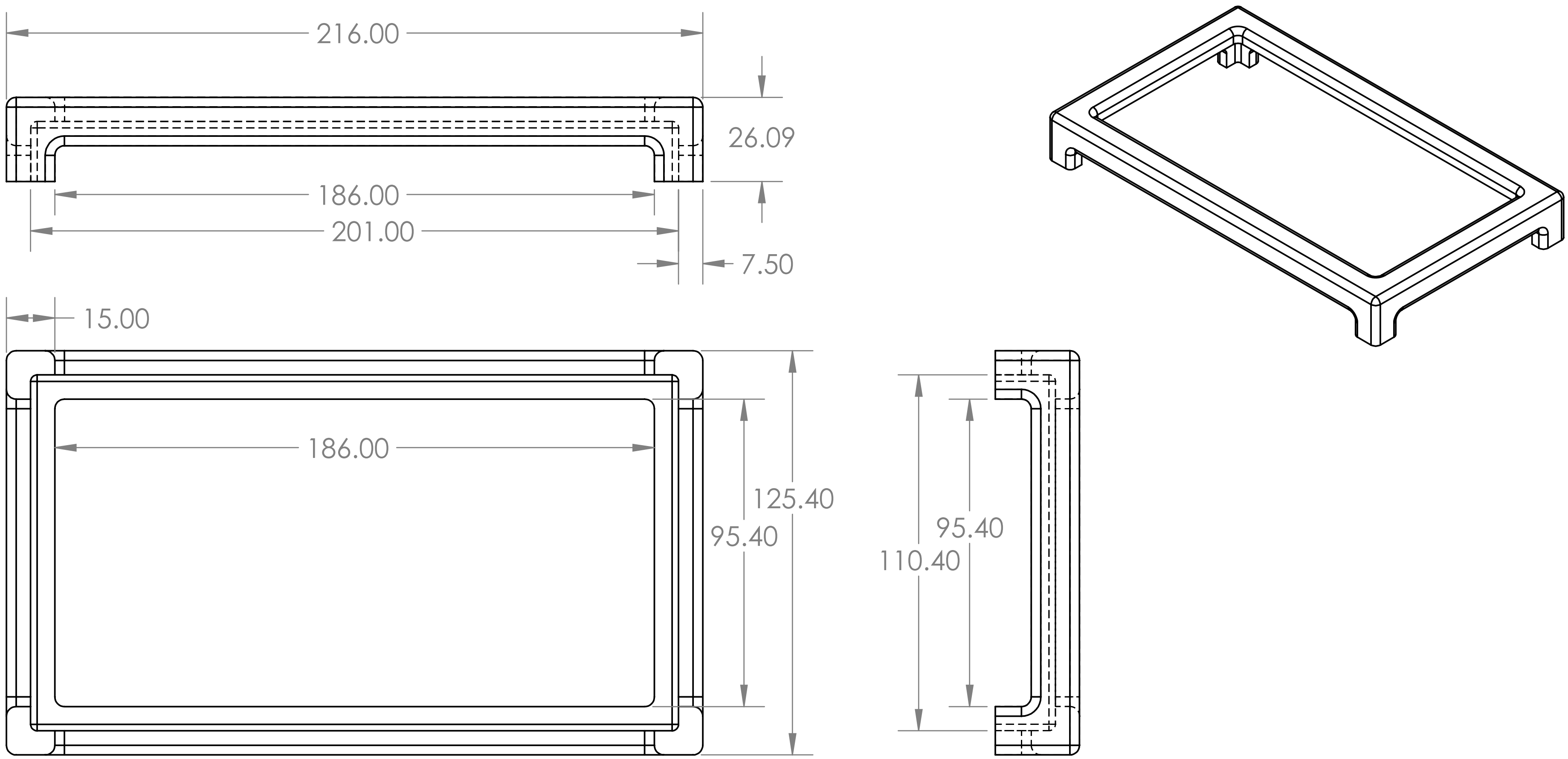
SCALE: 1:6 | WEIGHT: 951g | SHEET 1 OF 1



FDM 3D Printing

Print Time: 3.75 hr
on BambuLab A1

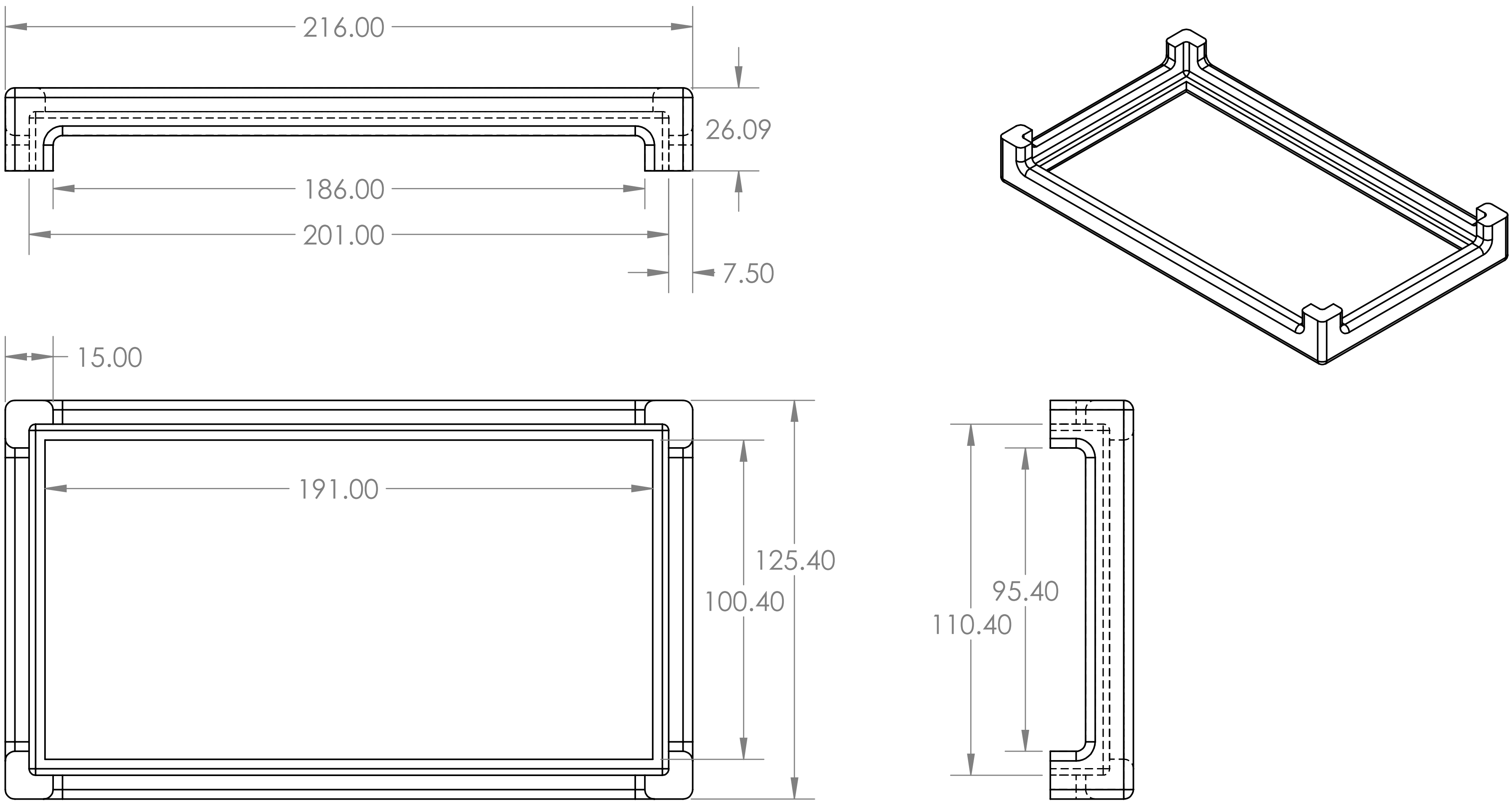
UNLESS OTHERWISE SPECIFIED:		NAME	DATE	
DIMENSIONS ARE IN MILLIMETERS TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	DRAWN	LAF	4/17/26	
	CHECKED			TITLE: Main Device Case
	ENG APPR.			
	MFG APPR.			
	Q.A.			
INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:			
MATERIAL PLA Plastic				SIZE A, DWG. NO., REV
FINISH				
DO NOT SCALE DRAWING				SCALE: 1:3, WEIGHT: 170g, SHEET 1 OF 1



3D Printing
Print time: 2.9 hours

All Fillets are 3mm

UNLESS OTHERWISE SPECIFIED:		NAME	DATE	
DIMENSIONS ARE IN MILLIMETERS TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	DRAWN	JK	4/17/26	
	CHECKED			TITLE:
	ENG APPR.			Bumper Top
	MFG APPR.			
INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			
MATERIAL TPU Plastic	COMMENTS:			SIZE A · DWG. NO. · REV
FINISH				
DO NOT SCALE DRAWING				SCALE: 1:2 · WEIGHT: 58.2g · SHEET 1 OF 1



3D Printing
Print time: 2.8 hours

All Fillets are 3mm

		NAME	DATE	
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DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	CHECKED			Bumper Bottom
	ENG APPR.			
	MFG APPR.			
INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			
MATERIAL TPU Plastic	COMMENTS:			SIZE A	DWG. NO.	REV
FINISH				
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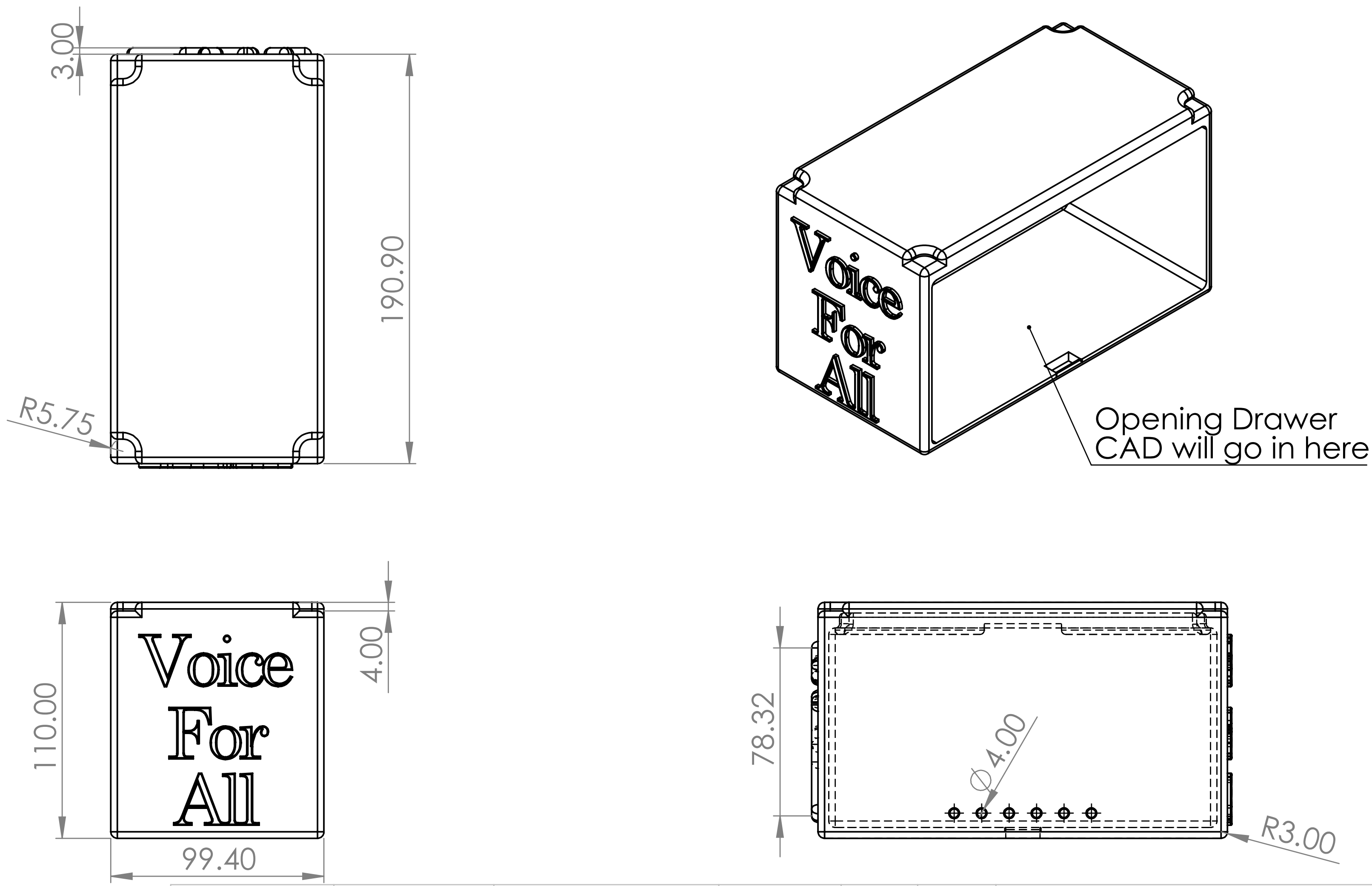
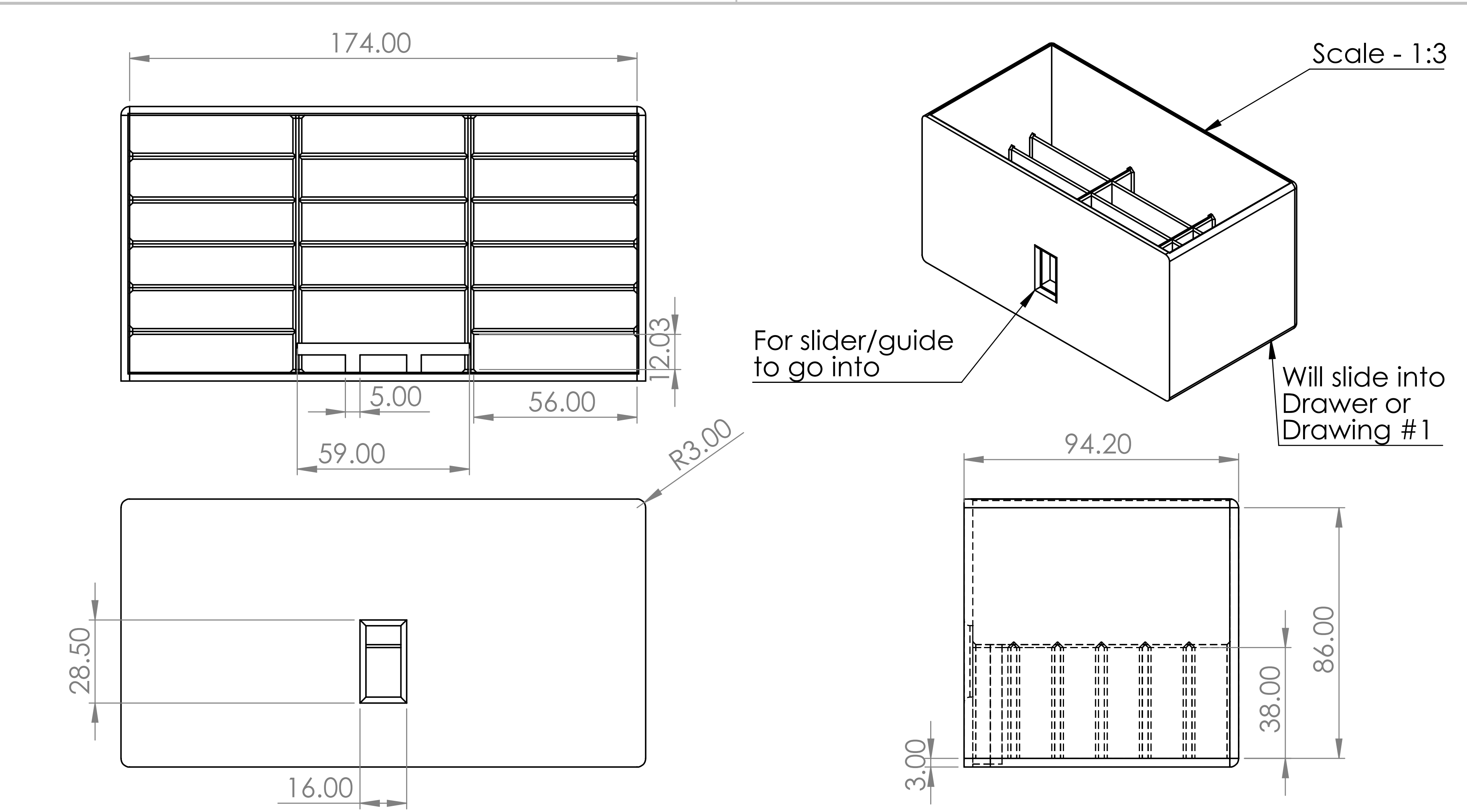


Manufacturing Type:
3D Printed
Material: PLA
Print Time: 7h 52 min
Weight: 290.65 g

				NAME	DATE	
		UNLESS OTHERWISE SPECIFIED:	DRAWN	Aiden M		
		DIMENSIONS ARE IN milimeters TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	CHECKED			TITLE: Drawer
			ENG APPR.			
			MFG APPR.			
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			
		MATERIAL	COMMENTS:			SIZE A / DWG. NO. 1 / REV
NEXT ASSY	USED ON	FINISH				
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:3 / WEIGHT: / SHEET 1 OF 1



Manufacturing Type:
3D Printed
Material: PLA
Print Time: 7h 20min
Weight: 310.47 g

UNLESS OTHERWISE SPECIFIED:

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TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL

FINISH

NEXT ASSY | USED ON

APPLICATION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	Aiden M	
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

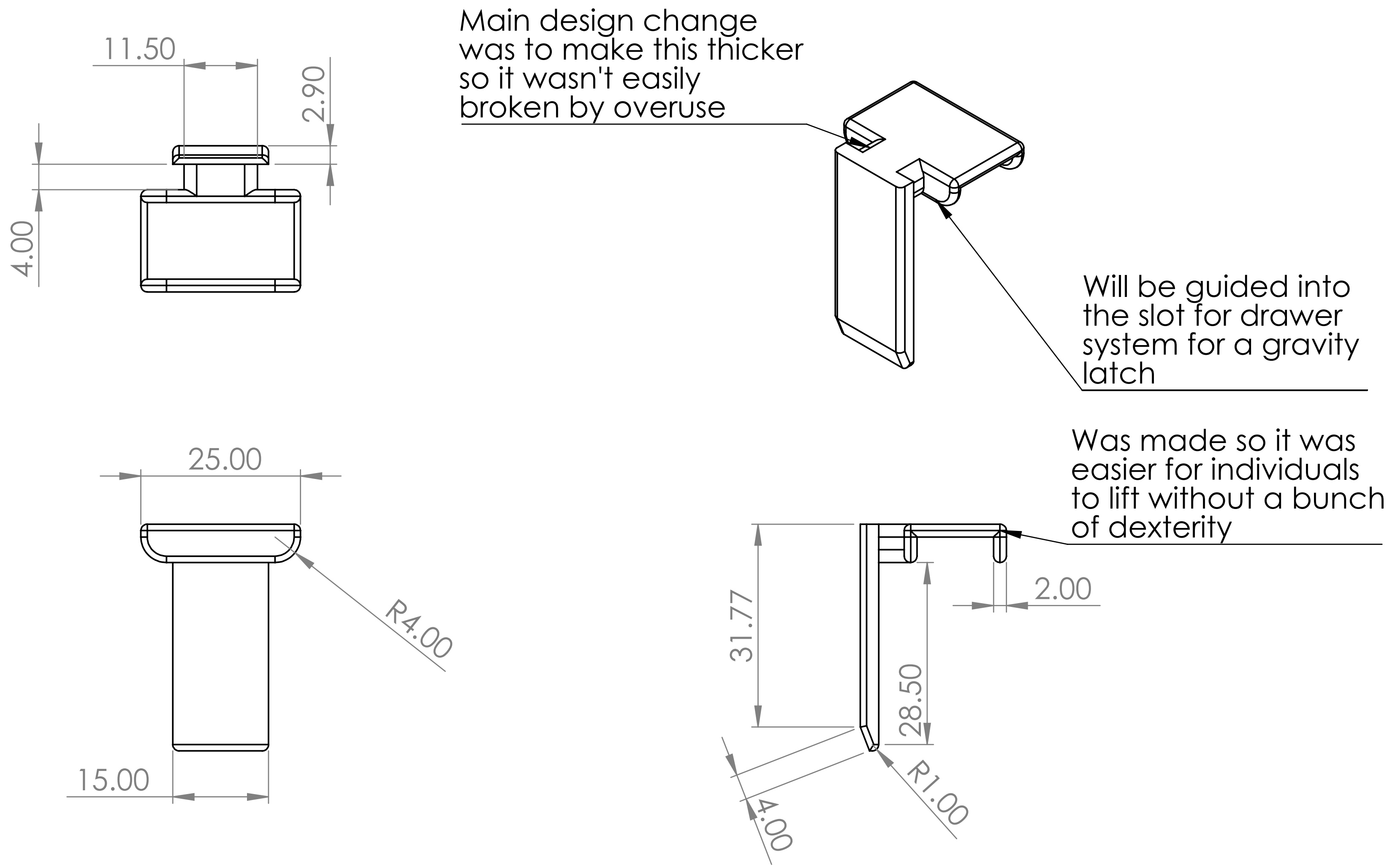
COMMENTS:

TITLE:

Opening Drawer

SIZE A | DWG. NO. 4 | REV

SCALE: 1:2 | WEIGHT: | SHEET 1 OF 1



Manufacturing Type:
3D Printed
Material: PLA
Print Time: 11m 8s
Weight: 3.41 g

UNLESS OTHERWISE SPECIFIED:		NAME	DATE		
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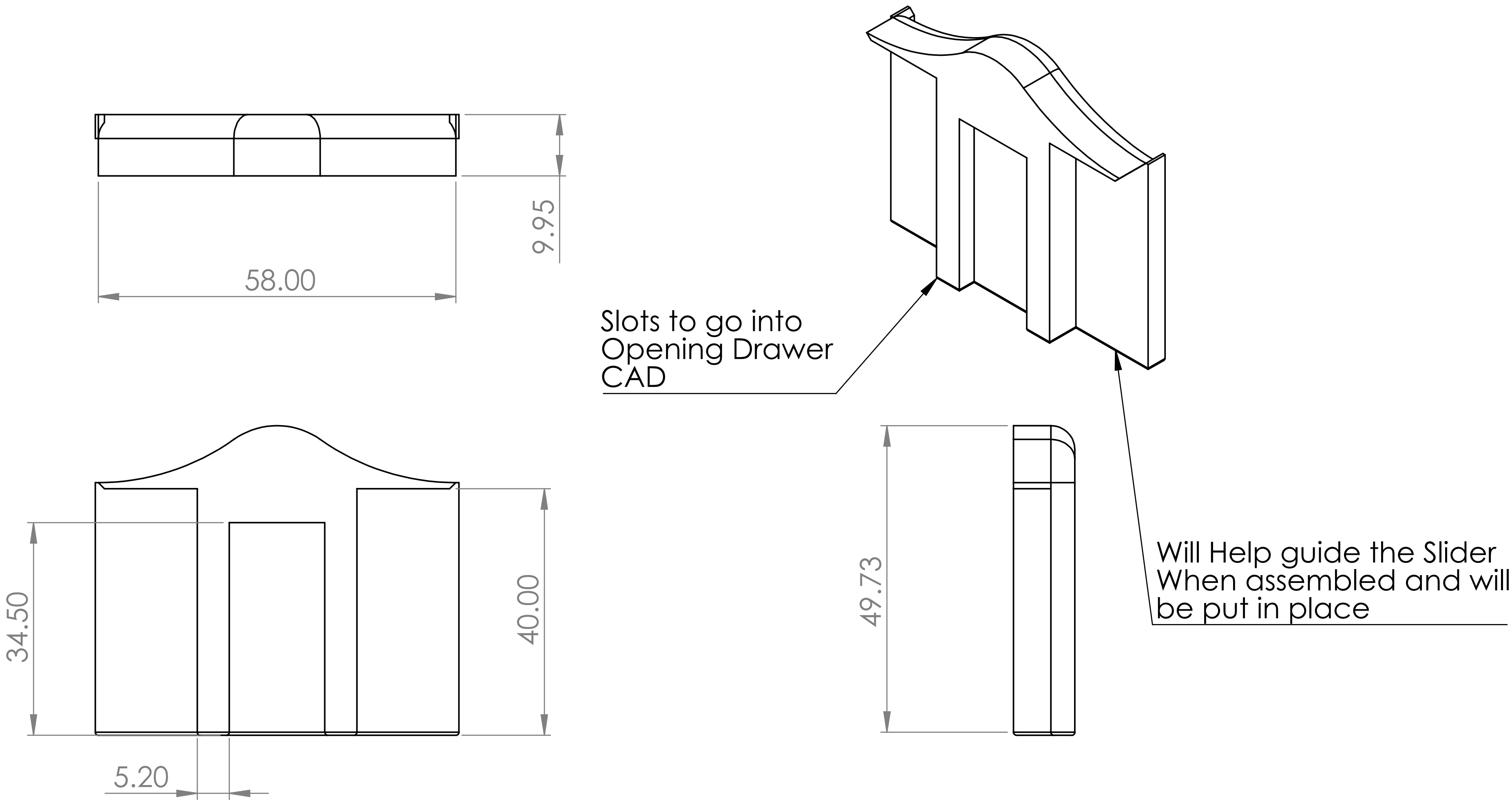
	CHECKED			Slider	
	ENG APPR.				
	MFG APPR.				
INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.				
MATERIAL	COMMENTS:			SIZE A	DWG. NO. 3
FINISH					REV
DO NOT SCALE DRAWING				SCALE: 1:1	WEIGHT: SHEET 1 OF 1

NEXT ASSY	USED ON
APPLICATION	



Manufacturing Type:
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Material: PLA
Print Time: 23min 26s
Weight: 11.1 g

NEXT ASSY	USED ON
APPLICATION	

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN Milimeters
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL

FINISH

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	Aiden M	
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

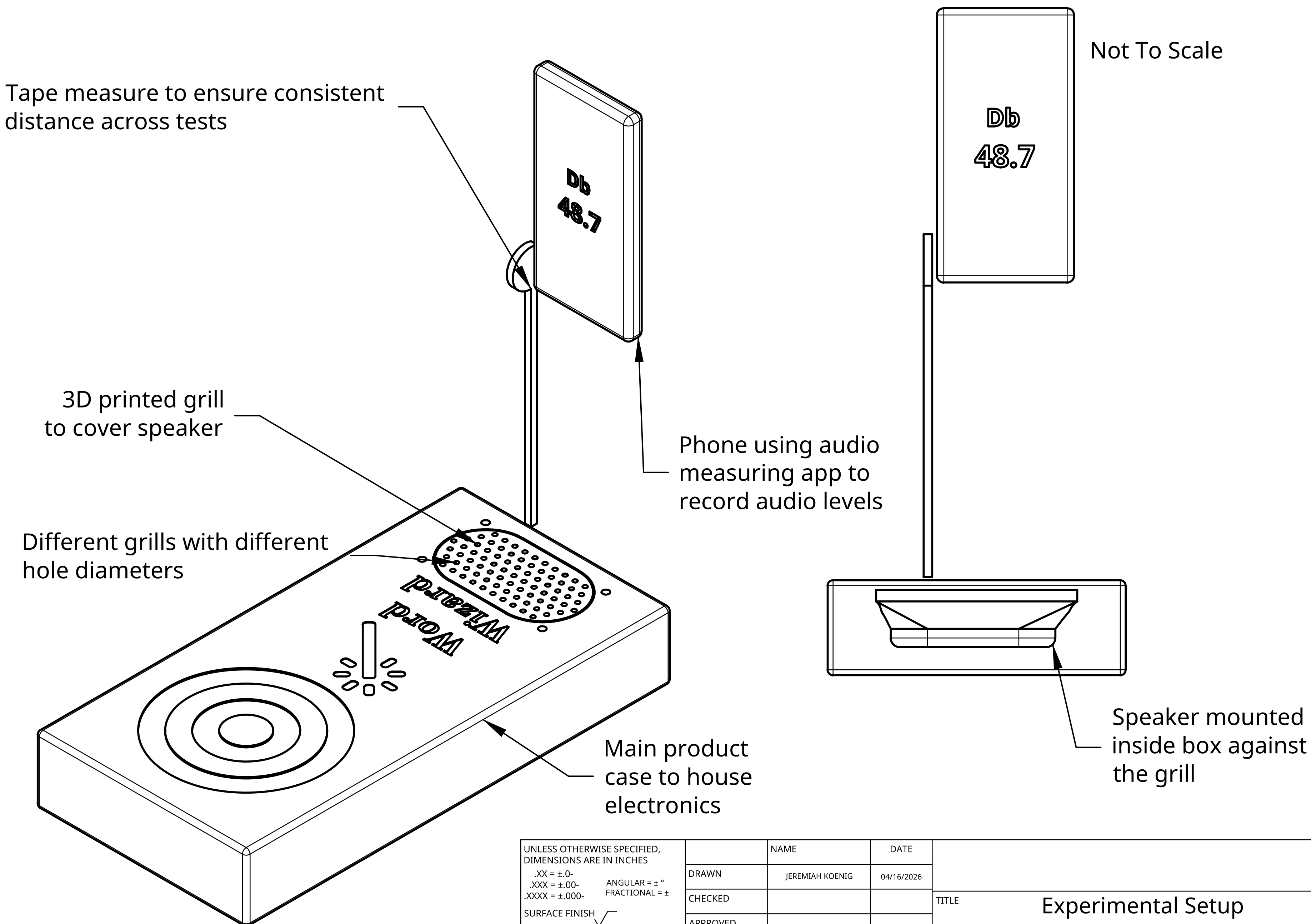
COMMENTS:

TITLE:

Slider Helper

SIZE	DWG. NO.	REV
A	2	

SCALE: 1:1 WEIGHT: SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH

DO NOT SCALE DRAWING

BREAK ALL SHARP EDGES AND REMOVE BURRS

THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	JEREMIAH KOENIG	04/16/2026
CHECKED		
APPROVED		

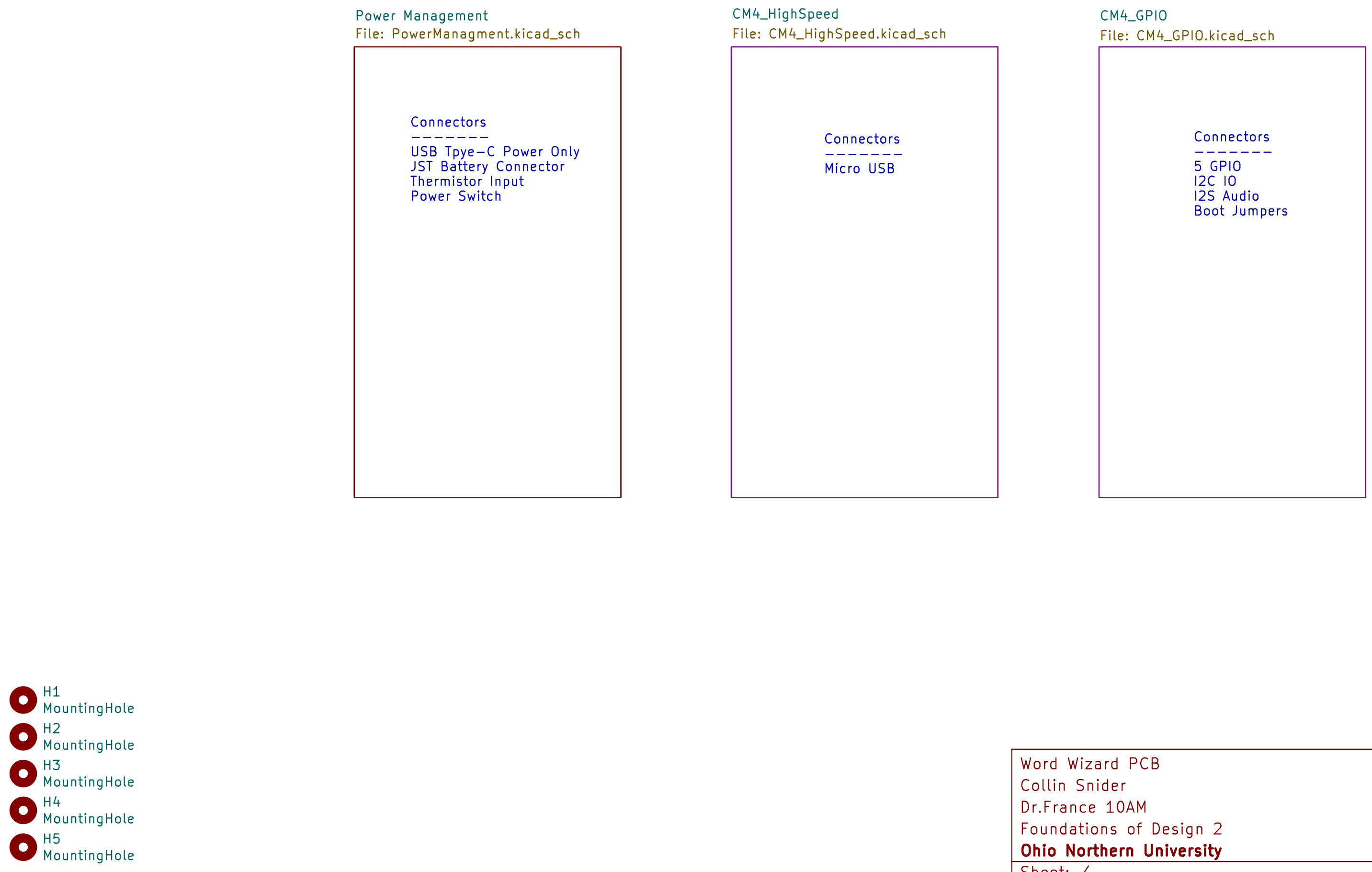
MATERIAL

FINISH

TITLE
Experimental Setup
10am Team 5

SIZE A | DWG NO. | REV.

SCALE NTS | WEIGHT | SHEET 1 of 1

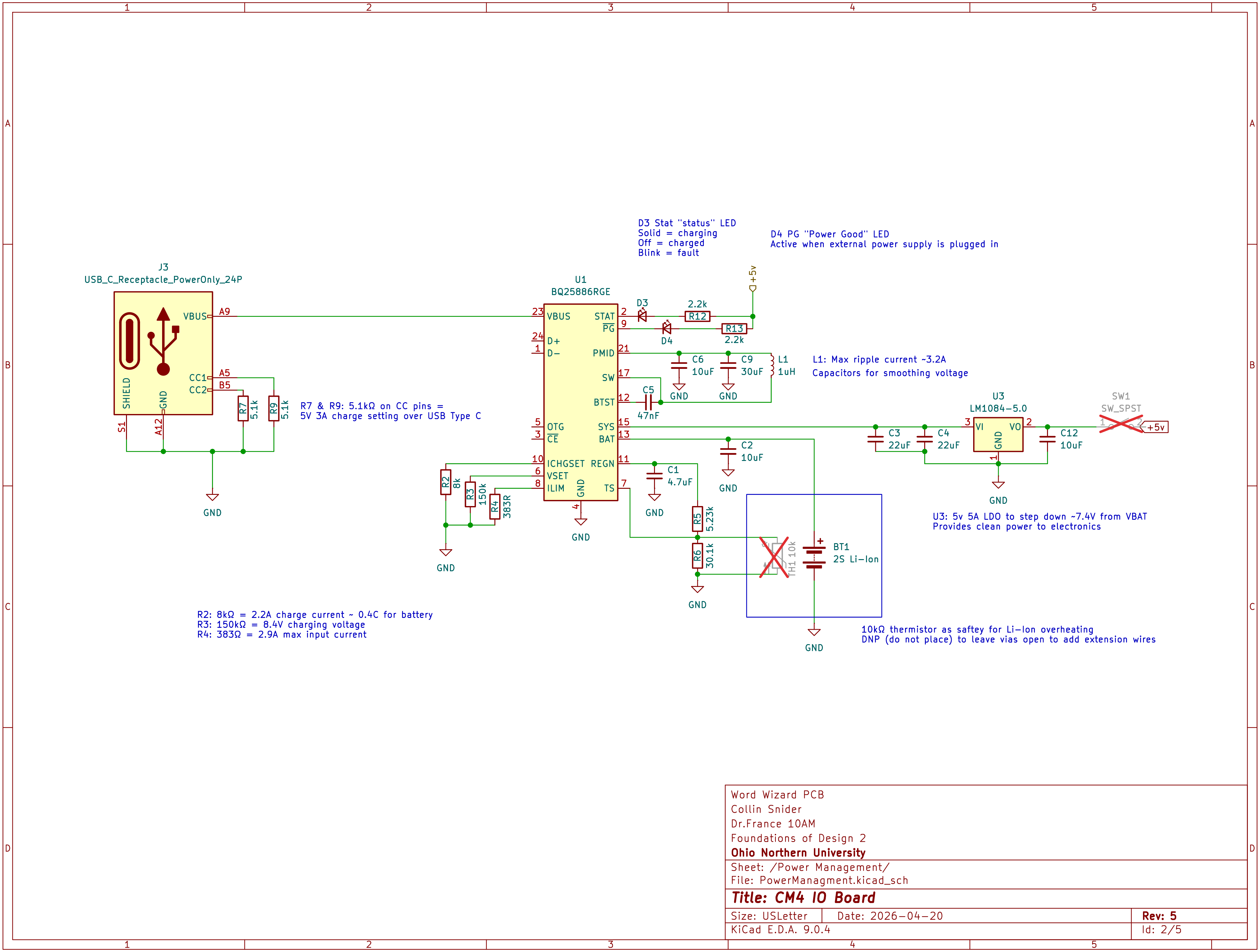


Word Wizard PCB
Collin Snider
Dr.France 10AM
Foundations of Design 2
Ohio Northern University

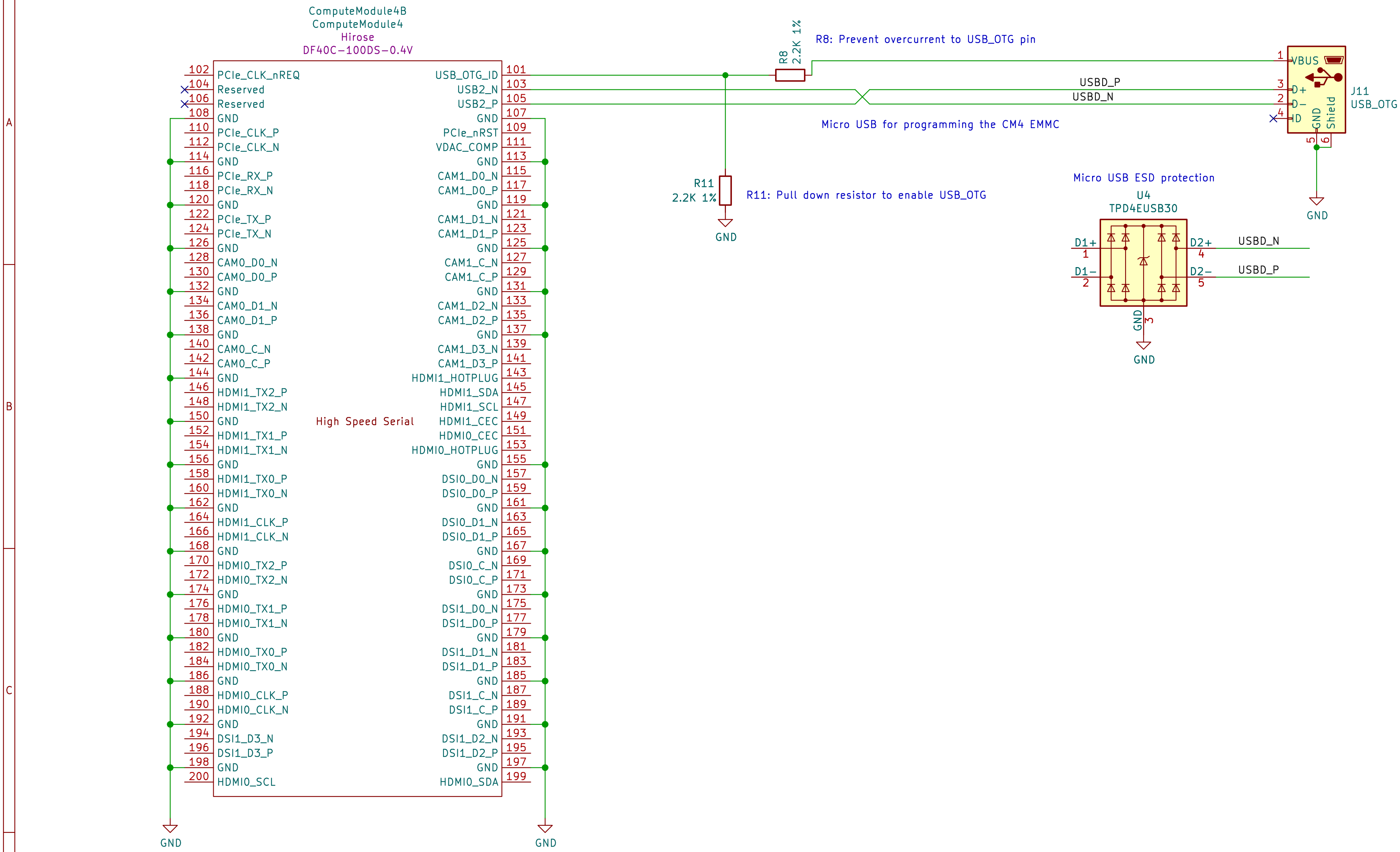
Sheet: /
File: Foundations of Design PCB R1.kicad_sch

Title: CM4 IO Board

Size: USLetter	Date: 2026-04-20	**Rev: 5**
KiCad E.D.A. 9.0.4		Id: 1/5



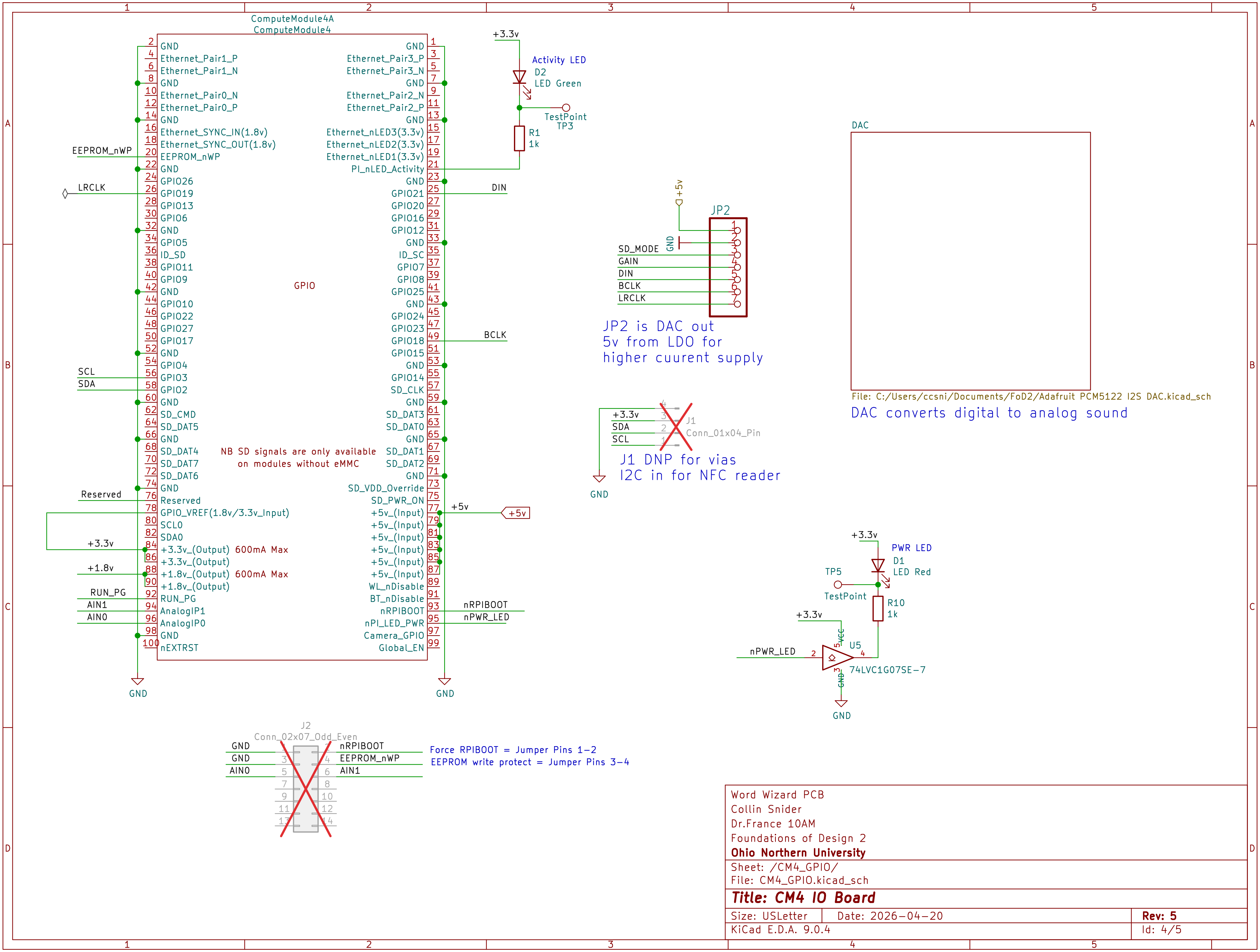
J3
USB_C_Receptacle_PowerOnly_24P
VBUS
A9
CC1
A5
CC2
B5
SHIELD
GND
S1
A12
R7
5.1k
R9
5.1k
GND
R7 & R9: 5.1kΩ on CC pins =
5V 3A charge setting over USB Type C
U1
BQ25886RGE
VBUS
D+
D−
OTG
CE
ICHGSET
VSET
ILIM
GND
STAT
PG
PMID
SW
BTST
SYS
BAT
REGN
TS
D3 Stat "status" LED
Solid = charging
Off = charged
Blink = fault
D4 PG "Power Good" LED
Active when external power supply is plugged in
+5v
D3
D4
R12
2.2k
R13
2.2k
C6
10uF
C9
30uF
L1
1uH
L1: Max ripple current ~3.2A
Capacitors for smoothing voltage
C5
47nF
C2
10uF
C1
4.7uF
R2
8k
R3
150k
R4
383R
R5
5.23k
R6
30.1k
TH1 10k
BT1
2S Li-Ion
U3
LM1084-5.0
VI
VO
C3
22uF
C4
22uF
C12
10uF
SW1
SW_SPST
U3: 5v 5A LDO to step down ~7.4V from VBAT
Provides clean power to electronics
10kΩ thermistor as saftey for Li-Ion overheating
DNP (do not place) to leave vias open to add extension wires
R2: 8kΩ = 2.2A charge current ~ 0.4C for battery
R3: 150kΩ = 8.4V charging voltage
R4: 383Ω = 2.9A max input current
Word Wizard PCB
Collin Snider
Dr.France 10AM
Foundations of Design 2
Ohio Northern University
Sheet: /Power Management/
File: PowerManagment.kicad_sch
Title: CM4 IO Board
Size: USLetter
Date: 2026-04-20
Rev: 5
KiCad E.D.A. 9.0.4
Id: 2/5

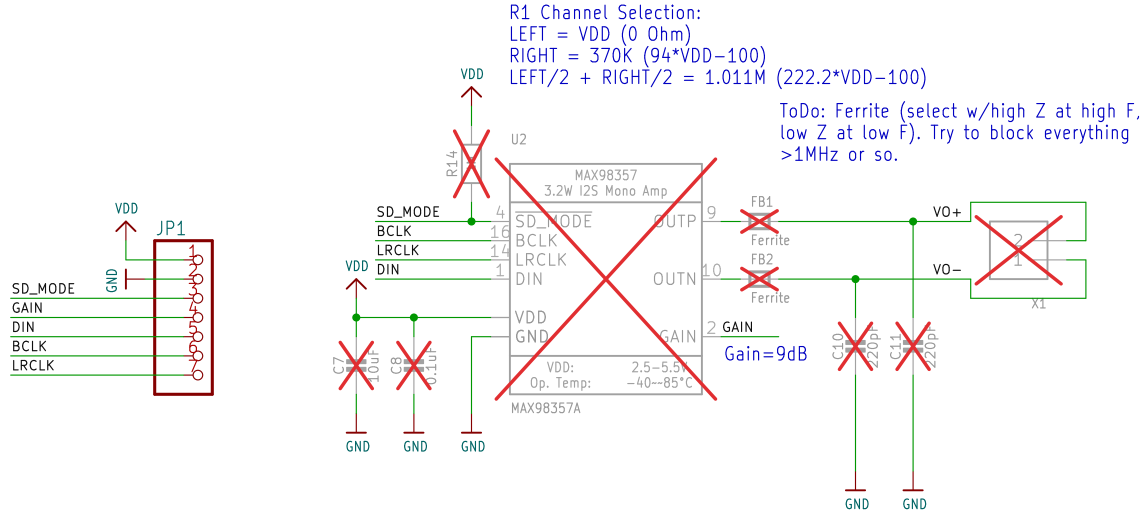


Word Wizard PCB
Collin Snider
Dr.France 10AM
Foundations of Design 2
Ohio Northern University
Sheet: /CM4_HighSpeed/
File: CM4_HighSpeed.kicad_sch

Title: CM4 IO Board

Size: USLetter	Date: 2026-04-20	**Rev: 5**
KiCad E.D.A. 9.0.4		Id: 3/5


ComputeModule4A
ComputeModule4
GPIO
2 GND
4 Ethernet_Pair1_P
6 Ethernet_Pair1_N
8 GND
10 Ethernet_Pair0_N
12 Ethernet_Pair0_P
14 GND
16 Ethernet_SYNC_IN(1.8v)
18 Ethernet_SYNC_OUT(1.8v)
20 EEPROM_nWP
22 GND
24 GPIO26
26 GPIO19
28 GPIO13
30 GPIO6
32 GND
34 GPIO5
36 ID_SD
38 GPIO11
40 GPIO9
42 GND
44 GPIO10
46 GPIO22
48 GPIO27
50 GPIO17
52 GND
54 GPIO4
56 GPIO3
58 GPIO2
60 GND
62 SD_CMD
64 SD_DAT5
66 GND
68 SD_DAT4
70 SD_DAT7
72 SD_DAT6
74 GND
76 Reserved
78 GPIO_VREF(1.8v/3.3v_Input)
80 SCL0
82 SDA0
84 +3.3v_(Output) 600mA Max
86 +3.3v_(Output)
88 +1.8v_(Output) 600mA Max
90 +1.8v_(Output)
92 RUN_PG
94 AnalogIP1
96 AnalogIP0
98 GND
100 nEXTRST
GND 1
Ethernet_Pair3_P 3
Ethernet_Pair3_N 5
GND 7
Ethernet_Pair2_N 9
Ethernet_Pair2_P 11
GND 13
Ethernet_nLED3(3.3v) 15
Ethernet_nLED2(3.3v) 17
Ethernet_nLED1(3.3v) 19
PI_nLED_Activity 21
GND 23
GPIO21 25
GPIO20 27
GPIO16 29
GPIO12 31
GND 33
ID_SC 35
GPIO7 37
GPIO8 39
GPIO25 41
GND 43
GPIO24 45
GPIO23 47
GPIO18 49
GPIO15 51
GND 53
GPIO14 55
SD_CLK 57
GND 59
SD_DAT3 61
SD_DAT0 63
GND 65
SD_DAT1 67
SD_DAT2 69
GND 71
SD_VDD_Override 73
SD_PWR_ON 75
+5v_(Input) 77
+5v_(Input) 79
+5v_(Input) 81
+5v_(Input) 83
+5v_(Input) 85
+5v_(Input) 87
WL_nDisable 89
BT_nDisable 91
nRPIBOOT 93
nPI_LED_PWR 95
Camera_GPIO 97
Global_EN 99
NB SD signals are only available
on modules without eMMC
EEPROM_nWP
LRCLK
SCL
SDA
Reserved
+3.3v
+1.8v
RUN_PG
AIN1
AIN0
GND
GND
DIN
BCLK
+5v
+5v
nRPIBOOT
nPWR_LED
+3.3v
Activity LED
D2
LED Green
TestPoint
TP3
R1
1k
+5v
JP2
GND
SD_MODE
GAIN
DIN
BCLK
LRCLK
1
2
3
4
5
6
7
JP2 is DAC out
5v from LDO for
higher cuurent supply
DAC
File: C:/Users/ccsni/Documents/FoD2/Adafruit PCM5122 I2S DAC.kicad_sch
DAC converts digital to analog sound
+3.3v
SDA
SCL
4
3
2
1
J1
Conn_01x04_Pin
GND
J1 DNP for vias
I2C in for NFC reader
+3.3v
PWR LED
D1
LED Red
TP5
TestPoint
R10
1k
+3.3v
VCC
U5
nPWR_LED
74LVC1G07SE-7
GND
GND
J2
Conn_02x07_Odd_Even
GND
GND
AIN0
nRPIBOOT
EEPROM_nWP
AIN1
Force RPIBOOT = Jumper Pins 1-2
EEPROM write protect = Jumper Pins 3-4
Word Wizard PCB
Collin Snider
Dr.France 10AM
Foundations of Design 2
Ohio Northern University
Sheet: /CM4_GPIO/
File: CM4_GPIO.kicad_sch
Title: CM4 IO Board
Size: USLetter
Date: 2026-04-20
Rev: 5
KiCad E.D.A. 9.0.4
Id: 4/5

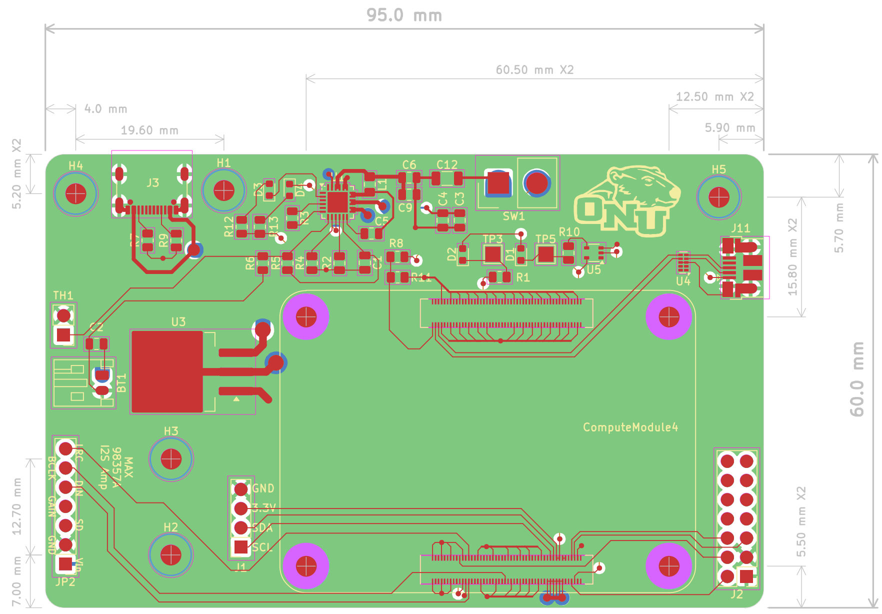


Word Wizard PCB
Collin Snider
Dr.France 10AM
Foundations of Design 2
Ohio Northern University

Sheet: /CM4_GPIO/DAC/
File: Adafruit PCM5122 I2S DAC.kicad_sch

Title: CM4 IO Board

Size: USLetter	Date: 2026-04-20	**Rev: 5**
KiCad E.D.A. 9.0.4		Id: 5/5



Word Wizard PCB Collin Snider Dr. France 10AM Foundations of Design 2 **Ohio Northern University**		
Sheet: File: Foundations of Design PCB R1.kicad_pcb		
Title: CM4 IO Board		
Size: USLetter	Date: 2026-04-20	**Rev: 5**
KiCad E.D.A. 9.0.4		Id: 1/1

Appendix 4: Budget Tracker

Item description	Purchased	Quantity Purchased	Vendor/source with link	Cost
NFC Cards	☐	20	Amazon	$8.99
Batteries	☐	1	Batteries.net	$38.39
3D Printing filament	☑		Donated	Donated
PCB	☑	5	JLCPCB	$22.79
NFC Card Reader	☐	1 (2 pack)	Amazon	$12.99
10 µF ±10% 16V Ceramic Capacitor	☑	2	DigiKey	$0.20
22 µF ±20% 10V Ceramic Capacitor	☑	2	DigiKey	$0.42
0.047 µF ±10% 50V Ceramic Capacitor	☑	1	DigiKey	$0.10
100 Position Connector Receptacle, Center Strip Contacts	☑	4	DigiKey	$7.56
Connector Header Through Hole, Right Angle 2 position	☑	2	DigiKey	$0.22
USB C charging port	☑	1	DigiKey	$0.52
1 µH Shielded Molded Inductor 3.2 A 50mOhm Max 0805	☑	1	DigiKey	$0.23
Red LED Indicator	☑	3	DigiKey	$0.36
Green LED Indicator	☑	1	DigiKey	$0.12
Lithium Ion / Polymer Charger	☑	1	DigiKey	$3.12
Buffer, Non-Inverting 1 Element 1 Bit per Element Open Drain	☑	1	DigiKey	$0.10
Linear Voltage Regulator IC Positive Fixed 1 Output 5A TO-263	☑	1	DigiKey	$2.97
RASPBERRY PI COMPUTE MOD	☑	1	DigiKey	$35.00
RASPBERRY PI BOARD	☑	1	DigiKey	$35.00
1-Channel (Mono) Output Class D Audio Amplifier Evaluation Board	☑	1	DigiKey	$5.95
SPEAKER	☑	1	DigiKey	$5.23
Tax and Shipping from DigiKey order	☑	1	PDF of DigiKey Order	$15.84
Tax and Shipping from Amazon order	☑	1		$1.59
Audio amp		1(2 pack)	Amazon	$7.38
		Expected Budget Remaining		**Expected Total Budget Spent**
		$4.93		$205.07

Appendix 5: Gantt Chart

Word Wizard - (Slogan - "Voice for All")

Team Members: Logan Frazer, Ethan Dsouza, Collin Snider, Jeremiah Koenig, Owen Willoby, Aiden McManis

#	Task	Who	Pre-Req	Start	End	Days	% Done
0	**Design Notebook**			23-Jan	2-Feb	11	100%
0.1	Create & share Google folder	LF		23-Jan	23-Jan	1	100%
0.1.1	Paste RFP in folder	LF	0.1	23-Jan	23-Jan	1	100%
0.1.2	Team Roster	LF	0.1	23-Jan	26-Jan	4	100%
0.1.3	Paste Meeting Agendas template in folder	LF	0.1	23-Jan	26-Jan	4	100%
0.2	Initial meeting	LF (lead)	0.1	27-Jan	27-Jan	1	100%
0.2.1	Team Contract	LF (lead)	0.2	27-Jan	27-Jan	1	100%
1	**Problem Framing + Conceptual Design**			23-Jan	16-Feb	25	100%
1.1	Project Focus Proposal Email	AM	0.2.1	3-Feb	5-Feb	3	100%
1.1.1	Opportunity Statement	AM	1.1	3-Feb	5-Feb	3	100%
1.2	Submit DR1 for TA's to Look over	ED	1. 1	8-Feb	9-Feb	2	100%
1.2.1	TA meeting for DR1	ED, CS, JK, OW, LF	1. 2	10-Feb	10-Feb	1	100%
1.2.2	Look over and make changes to DR1	JK	1.2.1	10-Feb	15-Feb	6	100%
1.2.3	Low Fi Model Completed	OW	1.2	10-Feb	15-Feb	6	100%
1.2.4	Design Sketch Completed	AM	1.2	10-Feb	15-Feb	6	100%
1.2.5	Gant Chart Completed	AM	1.2	10-Feb	15-Feb	6	100%
1.2.6	Bring Low-Fi and Design sketch to class	OW	1.2.4-5	13-Feb	13-Feb	1	100%
1. 3	Submit DR1	AM	1.2.5	15-Feb	15-Feb	1	100%
1.3.1	DR1 day	AM Remind All	1.3	16-Feb	16-Feb	1	100%
2	**Detailed Design**						
2.1	Talk about BMS to professionals/professors	CS	1.3	10-Feb	15-Feb	6	100%
2.1.1	Decide what to do with information	CS	2.1	15-Feb	17-Feb	3	100%
2.2	PCB ordered by Spring Break	ED	1.3	15-Feb	9-Mar	15	100%
2.3	Team Evaluation Survey	AM Remind All	N/A	23-Feb	23-Feb	1	100%
3	**Experimental Design**						
3.1	Experimentation set-up	OW	N/A	23-Feb	23-Feb	1	100%
3.1.1	Data Collection	OW	3.1	25-Feb	25-Feb	1	100%
3.1.2	Analysis	OW	3.1.1	27-Feb	27-Feb	1	100%
3.2	Individual CAD Drawings	AM Remind All	N/A	23-Feb	23-Mar	1	100%
3.2.1	TA Review of CAD Sign-up	ED	3.2	23-Feb	6-Mar	11	100%
3.2.2	Individual CAD Drawings (Print Out)	AM Remind All	3.2	23-Feb	23-Mar	28	100%
3.3	Register for Polar Pitch	AM remind All	N/A	20-Feb	20-Mar	1	100%
3.4	DR2 Submission	AM	3.2	20-Feb	23-Mar	32	100%
3.4.1	DR2 Day	AM Remind All	3.5	23-Mar	27-Mar	4	100%
4	**Comprehensive Design**						
4.1	Team Contract V2	LF	0.2.1	20-Feb	30-Mar	38	100%
4.2	DR3 Review	JK	3.4.1	23-Mar	20-Apr	29	100%
4.2.1	Value Propostion Video and Postcard	LF	N/A	1-Apr	15-Apr	14	100%
4.3	Comprehensive Design- Poster, Presentation, Prototype, and Paper	All	4.2.1	13-Apr	7-May	25	100%
4.3.1	Poster	OW & JK	4.2.1	13-Apr	6-May	23	100%
4.3.2	Presentation	CS & AM	4.2.1	13-Apr	7-May	24	100%
4.3.3	Prototype	All	4.2.1	13-Apr	7-May	24	100%
4.3.4	Paper	ED & LF	4.2.1	13-Apr	7-May	24	100%
4.4	DR3 Turn in	AM	4.2	23-Mar	20-Apr	29	100%
4.4.1	DR3 Day	AM Remind All	4.2.1	20-Apr	24-Apr	5	100%
4.5	Design Showcase	All	4.3	7-May	7-May	1	100%

Appendix 6: AI Disclosure

AI was used in problem framing, refining text, and programming the device. AI models used were Claude, Gemini, ChatGPT, and GPAI.

Appendix 7: Electronics Troubleshooting

Troubleshooting:

Unfortunately, but expectedly one of the most time consuming aspects of the project was attempting to diagnose an intermittent problem on the custom PCB. The code and hardware had previously been proven for functionality on a development board with jumpers, however upon transferring the compute module and components it began to encounter an error. For unknown reasons at the time, the board would function as expected on the surface level for about 20-30 seconds after boot up. However, when a card was introduced after this period it would lead to a boot cycle. The first diagnosing step taken was to isolate any potential power problems. Each board was given its own power supply and retested. Yet, the issue was not so simple. Next, the PN532 was hooked up to an oscilloscope to analyze its I2C data bus. See below.

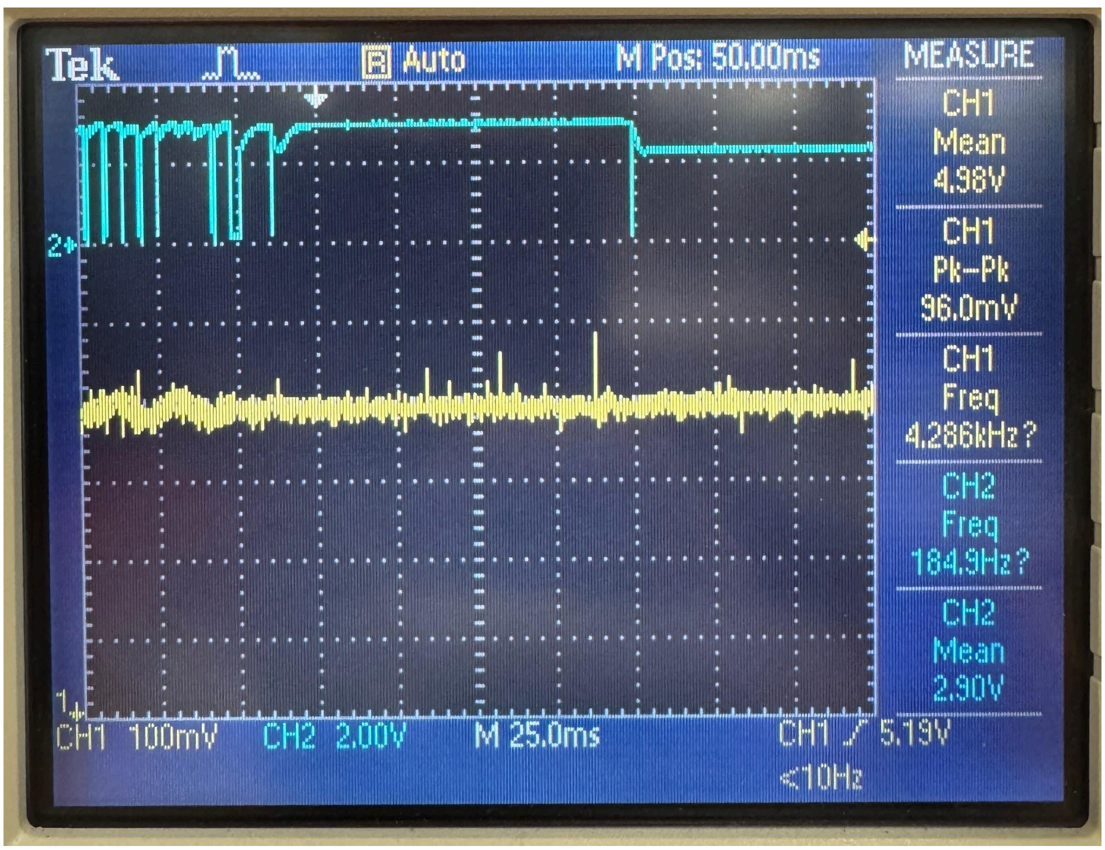


Figure 12: I2C communication, CH1 5V power, CH2 SDA

There are many details to note in this snapshot. First, we can note that the noise in the 5V rail is acceptable as it is well within the tolerance of ±5% or ±0.25V. Next, The I2C data line is not achieving full logic high of 3.3V. In the testing, it showed a maximum value of 3.1V. This is most concerning. It should be noted that there had been 2kΩ pull-up resistors to each I2C pin to achieve functionality. Since these pull-ups were in place, the logic high should always be 3.3V. In further reviewing of datasheets, these pull-ups were not necessary, as the board has internal 1.8kΩ pull-up resistors. However, the fact that the logic rail was not returning to 3.3V still suggested that something was ‘fighting’ the external pull up resistors. Our next hint comes from the failure mode of the I2C data pin. After about 20-30 seconds of abnormal operation, the data rail would return to 2.31V, once again despite the 3.3V pull-up resistors. This clearly shows that the board had detected some sort of fault and entered a fail-safe mode, powering down itself.

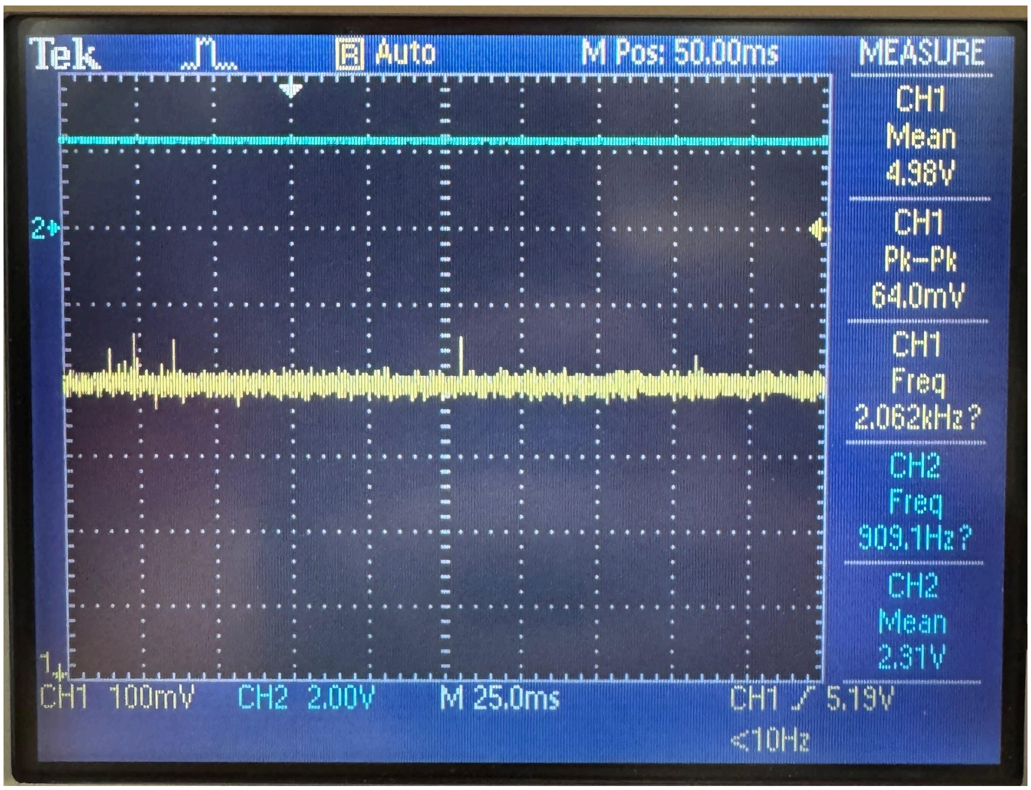


Figure 13: CM4 power shutoff, CH1 5V power, CH2 SDA

These specific fail cases led us to believe that something on the board was left ‘floating’ or not connected, as these issues did not occur with the exact same setup as the development board. After a complete review of development board schematics to ensure no circuitry was absent on the custom board, we went to review forums suggesting which pins left floating may cause such an issue. After much research, there were a few options to test, including enable or reset pins, however, GPIO_VREF turned out to be the pin that matched our exact symptoms. This input is the voltage reference which must be pulled directly to 1.8V or 3.3V to determine the logic level of the GPIO pins. Since it was left floating, it averaged to the mean between 0 and 5V, or approximately 2.3V. A simple overlook, and catastrophic consequences. This explains each problem we encountered, the external pull-ups, although enough to make the threshold of logic high, caused an extra thermal load in the GPIO controller, since 3.3V were ‘fighting’ the 2.5V reference. This explains how the board knew something was awry and reset itself by thermal protections, which realised about 20-30 seconds afterboot. The solution to this problem was to manually adjust the PCB traces and hardwire the voltage reference to 3.3V as it should have been designed.

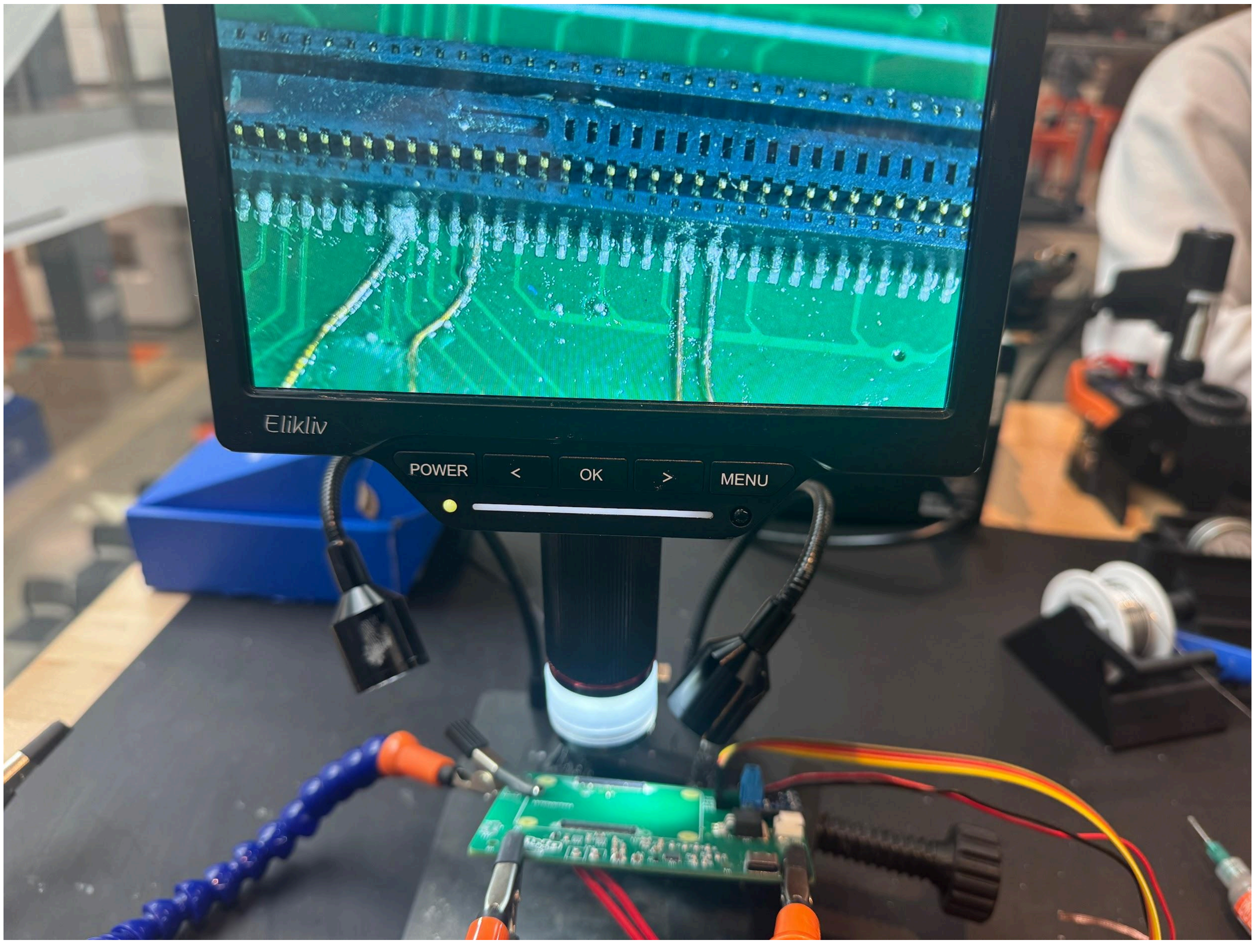


Figure 13: Manually adjusting PCB traces to Hirose connector

The above Figure 13, demonstrates the scale at which this repair had to be made, another previous adjustment can be seen. This was an attempt to rule out software errors, the I2C pins were swapped to ones that would match the development board to try to control as many variables as possible. On the next page, Figure 14, is a close up of the soldering process. Note that the wires pictured are 39 AWG or approximately 3 thousandths of an inch in diameter. A pair of 3.3V pins are leftmost, and the GPIO_ VREF is the adjacent solder connection. This process took a few hours alone to perfect. Too little or too much solder could lead to further complications.

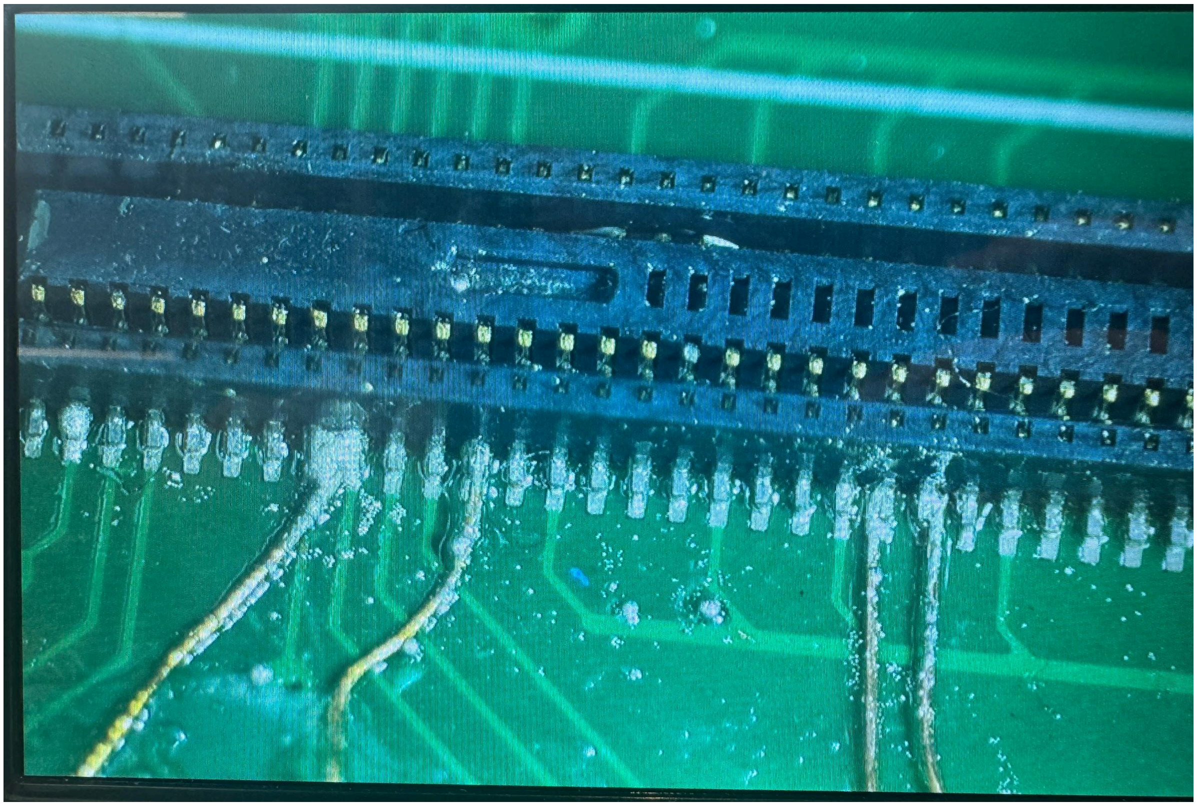


Figure 14: Soldering microscope view

The Next step was to make it back to the oscilloscope to verify the fix. Figure 15 demonstrates the setup for this last test and Figure 16 is a detailed view of the oscilloscope.

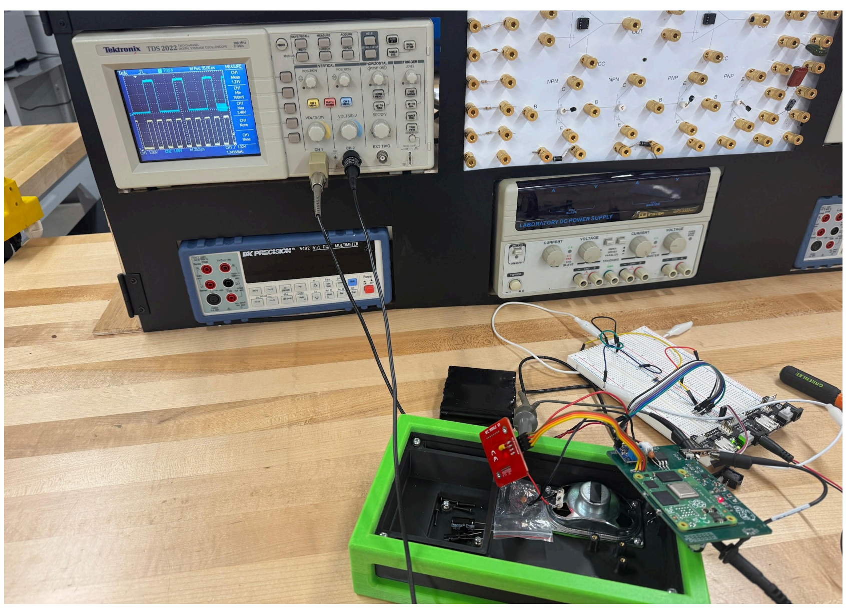


Figure 15: Diagnosing workbench setup

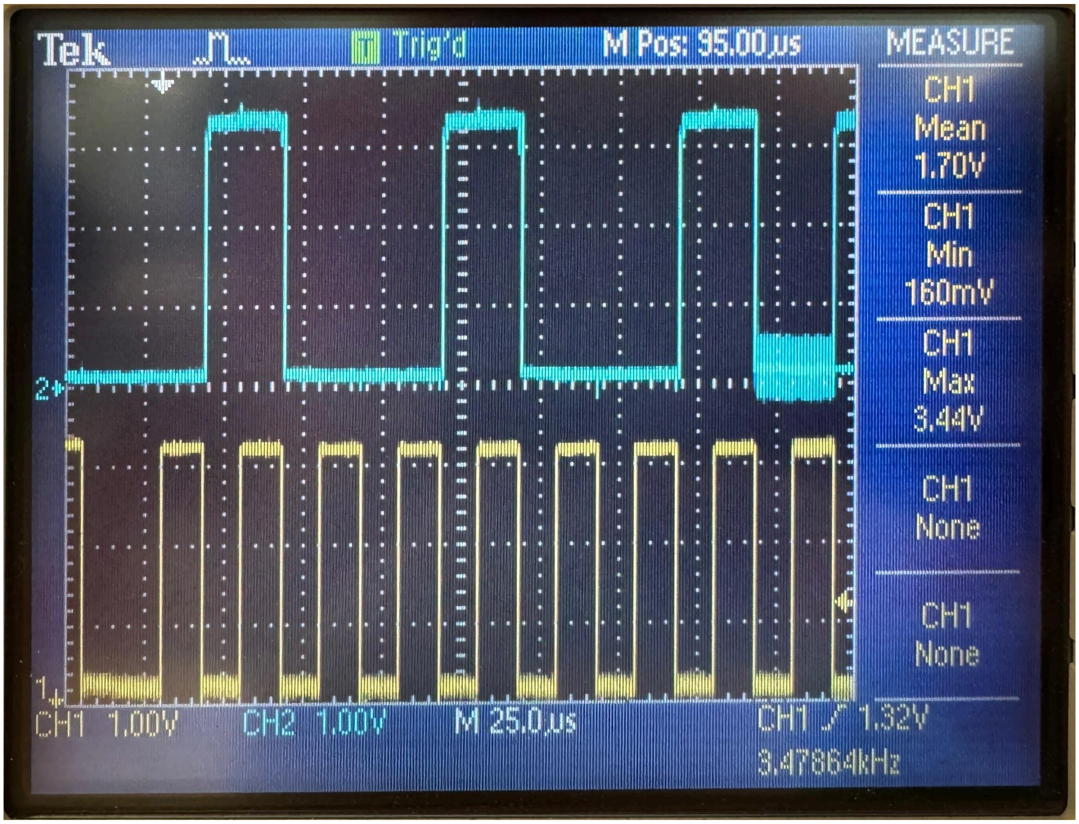


Figure 16: Fixed GPIO_VREF and digital communications. CH1 I2C SCL, CH2 I2C SDA

The results were exactly what we had hoped to see, clean rising and falling edges, well beyond the logic voltage thresholds. This overlook turned into a much harder to chase problem, however the lessons learned in the troubleshooting and diagnosing process are much greater than they would have been if everything had gone to plan. This problem turned out to be an invaluable teaching moment in both the art of measurement tools, soldering, but also emphasized the importance of checking, double checking, and peer reviewing designs whenever possible.

Appendix 8: User Manual

Word Wizard

Educator Reference Manual

Setup • Daily Operation • Care • Safety

Ohio Northern University – Team 5 | 2026

Introduction

Word Wizard is a portable, NFC-based augmentative and alternative communication (AAC) device designed to support non-verbal individuals in educational settings. When a communication card is tapped on the device, Word Wizard speaks the corresponding word or phrase aloud using a built-in text-to-speech (TTS) engine.

This manual covers everything a teacher or classroom staff member needs to set up, operate, maintain, and safely care for the device.

Safety Warnings

⚠️ **SCREWS / FASTENERS**
The enclosure uses accessible screws to allow component servicing. Do not allow unsupervised individuals to open the device. Loose screws are a choking and injury hazard.

⚠️ **BATTERY SAFETY**
Do not expose the device to extreme heat, puncture the enclosure, or submerge in liquid. Use any USB-C cable and a standard 5V charger. If the device becomes hot to the touch, smells unusual, or shows signs of swelling, discontinue use immediately.

⚠️ **DROP RISK**
Although the TPU bumper provides impact resistance for drops up to approximately 3 feet onto hard flooring, repeated or severe impacts can still damage internal components. Inspect the device after any significant drop.

What’s in the Box

- Word Wizard device (with TPU bumper attached)
- Communication card set (color-coded per grammatical group)
- Card storage drawer

📌 **NOTE**
Additional communication cards can be programmed using a smartphone with NFC write capability. See Section 5 for details.

Setup & First Use

Charging the Device

1. Connect the USB-C cable to the port on the back of the device.
2. Plug the other end into a standard USB wall adapter (5V recommended).
3. If powered on, green and orange status indicator lights will confirm charging is in progress. If the lights are blinking, disconnect charger and discontinue use.
4. Allow a full charge before first use. The battery capacity is at least 40 Watt-hours, providing extended daily use on a single charge.

> 📌 **TIP**
> Charge overnight before the first school day. We recommend charging each evening to ensure a full battery every morning.

Powering On

1. Toggle the power switch on the side.
2. The device will boot and be ready to use in less than a minute.
3. Place the device on a flat, stable surface accessible to the student.

Daily Operation

Using Communication Cards

Each communication card contains an embedded NFC tag. When a card is tapped on the top surface of the device, Word Wizard will speak the corresponding word or phrase aloud.

1. Hold the card flat and touch it to the NFC reader area, indicated by the magic wand logo.
2. The device will respond within a moment and speak the word.
3. Return the card to the storage drawer when not in use.

Card Color Coding (based on the Fitzgerald Key)

Cards follow the Fitzgerald Key color coding system, a widely recognized AAC standard that reduces cognitive load by grouping words by grammatical category:

Color	Category	Examples
Yellow	Question Words	why, where, how
Orange	Action Words	hungry, help, stop
Green	Social Words	yes, no, thank you
Pink	People	Mom, brother, grandma

Volume

The device is calibrated to output 60–65 dB at ear level, consistent with normal speech.

Language Support

Word Wizard uses the Piper TTS engine, which supports multiple languages and dialects. Contact the ONU Engineering Department if language or dialect settings need to be changed for a specific student.

Programming & Customizing Cards

One of the key advantages of Word Wizard is that communication cards can be customized or reprogrammed without purchasing new hardware. You will need a smartphone with NFC write capability (most modern Android and iOS devices support this).

1. Download a free NFC writing app on your smartphone (e.g., “NFC Tools” for iOS or Android).
2. Open the app and select “Write”, then “Add a Record”, then “Text”.
3. Enter the word or phrase you want the card to speak.
4. Select “Write / X Bytes” and hold the blank or existing NFC card against the back of your phone until the app confirms a successful write.
5. Tap the newly programmed card on Word Wizard to verify it works correctly.

📌 **NOTE**
NFC programming requires a smartphone capable of NFC write functionality. The device must be powered on to play a card, but does not need to be connected to a network. Cards can be reprogrammed as many times as needed.

Charging & Battery Care

- Use only standard USB-C cables rated for 5V charging.
- Avoid charging in direct sunlight or in environments above 35°C (95°F).
- Do not leave the device fully discharged for extended periods.
- A rechargeable internal battery means no disposable batteries are needed, reducing waste.
- If the device will not be used for more than one week, store it with a partial charge (around 50%).

> ⚠️ **DO NOT**
> Do not use non-standard or high-wattage fast chargers not rated for the device. Do not attempt to replace the battery yourself; contact the ONU Engineering Dept.

Cleaning & Hygiene

Word Wizard is designed with school hygiene in mind. The PLA enclosure and TPU bumper can be wiped down regularly without degradation.

Recommended Cleaning Method

1. Power off the device before cleaning.
2. Use a cloth lightly dampened with water and/or a gentle cleaning solution.
3. Avoid using alcohol to clean the acrylic back plate to reduce the risk of solvent stress cracking.
4. Wipe all exterior surfaces, including the bumper and NFC reader area.
5. Allow to dry fully before powering back on.
6. The communication cards can be wiped with the same method.

> ⚠️ **AVOID**
> Do not submerge the device in any liquid. Do not spray cleaners directly onto the device. Do not use abrasive materials or bleach-based cleaners, which may damage surface materials.

> 📌 **NOTE**
> Surface pores in 3D-printed parts may accumulate debris over time. Consistent wiping with cleaning solutions helps minimize bacterial buildup.

Communication Card Storage

The device includes an integrated card storage drawer at the base, organized by a modified Fitzgerald Key color coding. 40+ cards can be stored in the drawer.

- Keep cards sorted by color category for easy retrieval.
- Return cards to the drawer after each use to minimize loss.
- Cards should be stored flat to prevent bending or damage to the NFC chip.
- Consider labeling the drawer sections by color group if students manage cards independently.

> 📌 **TIP**
> Use the pictogram or image printed on the card face to help students identify and select cards independently, reducing reliance on reading ability.

Troubleshooting

Problem	Likely Cause	Solution
Device does not power on	Battery depleted	Charge device for at least 30 minutes before retrying.
Card tap produces no sound	Card not aligned / NFC issue	Re-tap the card flat and centered on the reader area. Try a different card to isolate the issue.
Sound is too quiet	Grill obstruction	Ensure speaker grill is clear of debris. Wipe grill with a dry cloth.
Card speaks the wrong word	Card was reprogrammed incorrectly	Re-write the card using the NFC programming steps in Section 5.
Device feels hot during charging	Charging environment too warm	Move device to a cooler location. If heat persists after moving, discontinue charging.
Device does not charge	Cable or port issue	Try a different USB-C cable or charger. Inspect port for debris and clear gently with a dry toothpick if needed.

📌 **STILL HAVING ISSUES?**
If the problem is not resolved by the steps above, do not attempt to open the device enclosure. Contact your building administrator or the team responsible for device maintenance.

Quick Reference Card

Cut out or print this section and post it near the device for easy access.

✅ Daily Startup Checklist	⚠️ Key Reminders
◻ Device charged overnight ◻ Power on–wait for “Powered On” vocal indicator ◻ Cards sorted in drawer by color ◻ Device placed within student’s reach ◻ Test one card tap before class begins	• USB-C only for charging • Wipe with damp cloth, never submerge • Do not open the enclosure • Report issues to ONU Engineering Dept.

Appendix 9: Code

Untitled-1

```
/**
 * Python script to read NFC card and push text data to Piper TTS
 * rpi_read_ndef_text.c
 *
 * Reads NDEF Text records from NFC cards (NTAG2xx / Mifare Ultralight)
 * using the pn532-lib on Raspberry Pi.
 *
 * Based on: https://github.com/soonuse/pn532-lib/blob/master/examples/raspberrypi/rpi_dump_ntag2.c
 *
 * Build:
 *   gcc -o rpi_read_ndef_text rpi_read_ndef_text.c -lpn532 -lpn532_rpi
 *
 * Wiring (I2C, default):
 *   PN532 SDA -> RPi GPIO 2 (Pin 3)
 *   PN532 SCL -> RPi GPIO 3 (Pin 5)
 *   PN532 VCC -> 3.3V or 5V
 *   PN532 GND -> GND
 */

#include <stdio.h>
#include <stdint.h>
#include <string.h>
#include "pn532.h"
#include "pn532_rpi.h"

/* ── NTAG2xx layout ──────────────────────────────────────────────────────────
 * Block 0-3  : UID / lock bytes / capability container
 * Block 4+   : user data (TLV-encoded NDEF messages)
 * Last blocks: dynamic lock / CFG (tag-size dependent)
 *
 * Each block is 4 bytes; total user blocks vary by tag variant:
 *   NTAG213 → blocks 4-39   (144 bytes)
 *   NTAG215 → blocks 4-129  (496 bytes)
 *   NTAG216 → blocks 4-225  (888 bytes)
 * We read up to block 135 (same ceiling as the dump example).
 * ──────────────────────────────────────────────────────────────────────────── */
```

```c

#define MAX_BLOCKS      135
#define BLOCK_SIZE        4
#define USER_START_BLOCK  4   /* first data block */

/* TLV tags */
#define TLV_NULL        0x00
#define TLV_NDEF        0x03
#define TLV_TERMINATOR  0xFE

/* NDEF TNF (Type Name Format) mask */
#define NDEF_TNF_MASK   0x07
#define NDEF_TNF_WELL_KNOWN 0x01

/* NDEF flags */
#define NDEF_FLAG_MB    0x80  /* Message Begin  */
#define NDEF_FLAG_ME    0x40  /* Message End    */
#define NDEF_FLAG_SR    0x10  /* Short Record   */

/* NDEF Well-Known Type: "T" = Text */
#define NDEF_TYPE_TEXT  'T'

/* ── Helper: print raw hex bytes ─────────────────────────────────────────── */
static void print_hex(const uint8_t *data, size_t len)
{
    for (size_t i = 0; i < len; i++) {
        printf("%02X ", data[i]);
    }
    printf("\n");
}

/* ── Parse and print one NDEF Text record payload ────────────────────────── */
/*
 * Text record payload format (RFC 5646):
 *   Byte 0       : status byte  – bit7=UTF16, bits5-0=lang length
 *   Bytes 1..L   : language code (e.g. "en")
 *   Bytes L+1..N : text (UTF-8 or UTF-16)
 */
```

```
static void parse_ndef_text(const uint8_t *payload, uint32_t payload_len)
{
    if (payload_len < 1) {
        printf("  [!] Payload too short\n");
        return;
    }

    uint8_t status    = payload[0];
    uint8_t lang_len  = status & 0x3F;        /* lower 6 bits */
    int     is_utf16  = (status & 0x80) != 0;

    if ((uint32_t)(1 + lang_len) >= payload_len) {
        printf("  [!] Malformed Text record\n");
        return;
    }

    /* Language code */
    char lang[64] = {0};
    memcpy(lang, payload + 1, lang_len);

    /* Text content */
    const uint8_t *text     = payload + 1 + lang_len;
    uint32_t       text_len = payload_len - 1 - lang_len;

    printf("  Encoding : %s\n",  is_utf16 ? "UTF-16" : "UTF-8");
    printf("  Language : %s\n",  lang);
    printf("  Text     : ");

    if (is_utf16) {
        /* Naïve UTF-16 LE print – strips high byte */
        for (uint32_t i = 0; i + 1 < text_len; i += 2) {
            putchar(text[i]);
        }
    } else {
        /* UTF-8: just print as-is (terminal must support UTF-8) */
        for (uint32_t i = 0; i < text_len; i++) {
            putchar(text[i]);
        }
```

```
    }
    putchar('\n');
}

/* ── Walk the TLV stream in the user-data buffer ────────────────────────── */
static void parse_ndef_tlv(const uint8_t *data, size_t data_len)
{
    size_t  pos       = 0;
    int     found_any  = 0;

    while (pos < data_len) {
        uint8_t tag = data[pos++];

        if (tag == TLV_NULL)       { continue; }
        if (tag == TLV_TERMINATOR) { break;    }

        if (pos >= data_len) break;

        /* TLV length – extended (3-byte) form if 0xFF */
        uint32_t length;
        if (data[pos] == 0xFF) {
            if (pos + 2 >= data_len) break;
            length = ((uint32_t)data[pos + 1] << 8) | data[pos + 2];
            pos += 3;
        } else {
            length = data[pos++];
        }

        if (pos + length > data_len) {
            printf("[!] TLV value exceeds buffer – tag 0x%02X, len %u\n",
                   tag, length);
            break;
        }

        if (tag != TLV_NDEF) {
            pos += length; /* skip unknown TLV */
            continue;
        }
```

```
        /* ── Found an NDEF message TLV ─────────────────────────── */
        const uint8_t *ndef_msg    = data + pos;
        size_t         ndef_remain = length;
        pos += length;

        /* Walk individual NDEF records inside the message */
        size_t rpos = 0;
        while (rpos < ndef_remain) {
            if (rpos + 3 > ndef_remain) break;

            uint8_t flags    = ndef_msg[rpos++];
            uint8_t type_len = ndef_msg[rpos++];

            /* Payload length: SR flag → 1 byte, else 4 bytes */
            uint32_t payload_len;
            if (flags & NDEF_FLAG_SR) {
                if (rpos >= ndef_remain) break;
                payload_len = ndef_msg[rpos++];
            } else {
                if (rpos + 4 > ndef_remain) break;
                payload_len = ((uint32_t)ndef_msg[rpos    ] << 24) |
                              ((uint32_t)ndef_msg[rpos + 1] << 16) |
                              ((uint32_t)ndef_msg[rpos + 2] <<  8) |
                               (uint32_t)ndef_msg[rpos + 3];
                rpos += 4;
            }

            /* ID length (if IL flag set) */
            uint8_t id_len = 0;
            if (flags & 0x08) { /* IL flag */
                if (rpos >= ndef_remain) break;
                id_len = ndef_msg[rpos++];
            }

            if (rpos + type_len + id_len + payload_len > ndef_remain) break;

            const uint8_t *type    = ndef_msg + rpos; rpos += type_len;
```

```c
            /* skip ID */                              rpos += id_len;
            const uint8_t *payload = ndef_msg + rpos; rpos += payload_len;

            uint8_t tnf = flags & NDEF_TNF_MASK;

            printf("\n[NDEF Record]\n");
            printf("  TNF      : 0x%02X (%s)\n", tnf,
                   tnf == NDEF_TNF_WELL_KNOWN ? "Well-Known" : "other");

            if (type_len > 0) {
                printf("  Type     : %c\n", type[0]);
            }

            if (tnf == NDEF_TNF_WELL_KNOWN &&
                type_len == 1 && type[0] == NDEF_TYPE_TEXT) {
                parse_ndef_text(payload, payload_len);
                found_any = 1;
            } else {
                printf("  Payload  : ");
                print_hex(payload, payload_len);
            }

            if (flags & NDEF_FLAG_ME) break; /* last record in message */
        }
    }

    if (!found_any) {
        printf("[i] No NDEF Text records found on this card.\n");
    }
}

/* ── main ────────────────────────────────────────────────────────────────── */
int main(void)
{
    uint8_t  block_buf[BLOCK_SIZE];
    uint8_t  uid[MIFARE_UID_MAX_LENGTH];
    uint8_t  user_data[MAX_BLOCKS * BLOCK_SIZE];
    uint32_t user_data_len = 0;
```

```
    printf("=== PN532 NDEF Text Reader ===\n\n");

    PN532 pn532;
    /* Choose ONE init method to match your hardware wiring: */
    PN532_I2C_Init(&pn532);
    /* PN532_SPI_Init(&pn532);  */
    /* PN532_UART_Init(&pn532); */

    uint8_t fw[4];
    if (PN532_GetFirmwareVersion(&pn532, fw) != PN532_STATUS_OK) {
        fprintf(stderr, "[!] PN532 not found. Check wiring.\n");
        return -1;
    }
    printf("[+] PN532 detected – firmware v%d.%d\n\n", fw[1], fw[2]);

    PN532_SamConfiguration(&pn532);

    printf("Waiting for NFC card...\n");

    /* ── Poll until a card is presented ──────────────────────────────── */
    int32_t uid_len;
    while (1) {
        uid_len = PN532_ReadPassiveTarget(&pn532, uid,
                                          PN532_MIFARE_ISO14443A, 1000);
        if (uid_len != PN532_STATUS_ERROR) break;
        printf(".");
        fflush(stdout);
    }

    printf("\n[+] Card detected!\n");
    printf("    UID: ");
    for (int32_t i = 0; i < uid_len; i++) {
        printf("%02X ", uid[i]);
    }
    printf("(%d bytes)\n\n", uid_len);

    /* ── Read user data blocks ──────────────────────────────────────── */
```

```
    printf("Reading data blocks...\n");
    for (uint8_t blk = USER_START_BLOCK; blk < MAX_BLOCKS; blk++) {
        uint32_t err = PN532_Ntag2xxReadBlock(&pn532, block_buf, blk);
        if (err != PN532_ERROR_NONE) {
            /* End of readable area (e.g. hit config pages) */
            break;
        }
        memcpy(user_data + user_data_len, block_buf, BLOCK_SIZE);
        user_data_len += BLOCK_SIZE;
    }

    printf("[+] Read %u bytes of user data.\n\n", user_data_len);

    /* ── Parse TLV / NDEF ─────────────────────────────────────── */
    printf("Parsing NDEF records...\n");
    parse_ndef_tlv(user_data, user_data_len);

    printf("\nDone.\n");
    return 0;
}
```

References

DR1

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DR2

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DR3

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Individual Contributions

Table 7: Individual contributions

Name	Contributions	Hours since DR3
Ethan D'Souza	1. created opportunity statement and key design features collage in final paper 2. took photos for final paper 3. Wrote Next Steps paragraph in final paper 4. Formatted and edited problem framing documentation throughout semester and formatted in final paper 5. updated and formatted testplan in final paper 6. updated and formatted references section in final paper 7. Completed Eco-Design Strategies paragraph for final paper 8. Completed Broader Impacts section for final paper a. researched and completed benefits and risks/tradeoffs 9. Completed formatting and cleanliness of final paper 10. contributed to revision of Poster and completed electronics selection	6
Logan Frazer	1. Helped with poster content and formatting 2. Helped with paper content and formatting	6

	3. Programmed NFC cards with final word set 4. Updated and finalized CAD drawings 5. Updated figures 1-5 6. Annotated main case assembly drawing	
Aiden McManis	1. Helped with Poster with organization, and to make sure everything looked good 2. Gantt Chart put into doc 3. Helped with presentation for Design Showcase	4
Owen Willoby	1. Created the Poster and worked to finalize it 2. Kept and organized all receipts and finalized the budget 3. Helped with presentation for design showcase 4. Helped troubleshoot product and fix product 5. Helped with paper content and formatting	5
Jeremiah Koenig	1. Helped finish product (clean up all parts, put final components on PCB, assemble final product) 2. Helped touch up poster 3. Helped work on presentation	10
Collin Snider	1. PCB diagnosing and repair 2. Install new board 3. Helped with poster design 4. Helped with final paper 5. Helped to assemble and clean up wiring of electronics in the final product	25